INCH-POUND
MIL-DTL-15730N(SH)
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SUPERSEDING
MIL-DTL-15730N(SH)
9 June 2021

DETAIL SPECIFICATION

COOLERS, FLUID, NAVAL SHIPBOARD – HYDROCARBON BASE OIL AND FRESHWATER SERVICES

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers naval shipboard, fluid coolers for use in hydrocarbon base oil and freshwater services for which standards of resistance to shock and standards of construction are higher than for usual commercial marine use.

1.2 Classification. Coolers are of the following types and classes as specified (see 6.2).

1.2.1 Type I. Type I coolers are of the shell and tube design with the coolant circulated through the tubes and the fluid to be cooled passed through the shell.

a. Class 1 – Submarine, seawater cooled, seawater side subject to sea pressure for greater than 200-foot submergence, cyclic life stipulated.

b. Class 2 – Submarine, seawater cooled, seawater side secured for greater than 200-foot submergence.

c. Class 3 – Surface ship, seawater cooled.

d. Class 4 – Submarine, freshwater cooled.

e. Class 5 – Surface ship, freshwater cooled.

1.2.2 Type II. Type II coolers are of the fabricated tube design with the coolant circulated through the casing and the fluid to be cooled passed through the tubes.

a. Class 2 – Submarine, seawater cooled, seawater side secured for greater than 200-foot submergence.

b. Class 3 – Surface ship, seawater cooled.

c. Class 4 – Submarine, freshwater cooled.

d. Class 5 – Surface ship, freshwater cooled.

1.3 Application of cooler types and classes. Intended cooler application will be as specified (see 6.2) and subject to the following considerations:

a. Coolers for submarine service with seawater as the coolant that will be subject to sea pressure at submergence greater than 200 feet are type I, class 1. For these coolers, design and analysis are as specified in 3.3.15.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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b. Coolers for submarine service with seawater as the coolant for applications in which the seawater side of the cooler will be secured for submergence greater than 200 feet are type I, class 2.

c. Coolers for surface ships and service craft with seawater as the coolant are type I or II, class 3.

d. Coolers for submarine service with fresh water as the coolant are type I or II, class 4.

e. Coolers for surface ship and service craft with fresh water as the coolant are type I or II, class 5.

f. Lubricating oil coolers for main propulsion turbines and gears or for main propulsion gears are type I.

g. Lubricating oil coolers using scoop injection of circulating water are of single-pass construction on the coolant-side and are straight tube, type I.

h. Hydraulic oil coolers are type I.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-C-450 - Copper-Aluminum Alloy (Aluminum Bronze) Plate, Sheet, Strip, and Bar (Copper Alloy Numbers 606, 610, 613, 614, and 630)

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-59004 - Anti-Galling Compound, Thread Lubricating, Seizing Resistant, and Calcium Hydroxide Containing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-PRF-1149 - Gasket Materials, Synthetic Rubber, 50 and 65 Durometer Hardness

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for

MIL-PRF-6855 - Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification for

MIL-T-15005 - Tubes, Condenser and Heat Exchanger, Copper-Nickel Alloys (UNS C70600 & C71500)

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

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MIL-P-15024/5	-	Plates, Identification
MIL-C-15726	-	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod and Wire
MIL-E-15809	-	Expander Tube Condenser and Heat Exchangers
MIL-T-16420	-	Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-DTL-18001	-	Anodes, Sacrificial Zinc Alloy
MIL-A-19521	-	Anode Retaining Support Plugs and Anode Selection and Installation Design Criteria for Shipboard Condensers and Heat Exchangers
MIL-PRF-20042	-	Flanges, Pipe, Bronze (Silver Brazing)
MIL-G-21610	-	Gaskets, Heat Exchanger, Various Cross Section Rings, Synthetic Rubber
MIL-T-22214	-	Tube, Condenser and Heat Exchanger With Integral Fins (UNS Alloy Nos. C71500, C70600, C12200)
MIL-S-22698	-	Steel Plate, Shapes and Bars, Weldable Ordinary Strength and Higher Strength: Structural
MIL-DTL-24106	-	Nickel-Copper Alloy Bars, Rods, and Forgings (UNS N04400)
MIL-T-24107	-	Tube, Copper (Seamless) (Copper Alloy Numbers C10100, C10200, C10300, C10800, C12000, C12200, and C14200)
MIL-L-24478	-	Lubricant, Molybdenum Disulfide in Isopropanol
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service
MIL-P-24691/1	-	Pipe and Tube, Carbon Steel, Seamless
MIL-DTL-24696	-	Gasket, Sheet, Non-Asbestos
MIL-DTL-24707/1	-	Castings, Ferrous, for Machinery and Structural Applications

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22	-	Welded Joint Design
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-438	-	Schedule of Piping, Valves, Fittings, & Associated Piping Components for Submarines Service NOTE: MIL-STD-438 is inactive for new design and intended only for guidance. It will eventually be replaced.
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
DOD-STD-1399-301	-	Interface Standard for Shipboard Systems, Section 301A, Ship Motion and Attitude (Metric)
MIL-STD-2193	-	Ship Hydraulic System Components

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-5959186 - Anode Plug, Submarine Heat Exchanger

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an account for drawing access, send an email to nnsy_jedmics_nsedr_helpdesk@us.navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0900-LP-001-7000	-	Fabrication and Inspection of Brazed Piping Systems
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	-	Unified Inch Screw Threads (UN, UNR, and UNJ Thread Forms)
ASME B1.12	-	Class 5 Interference-Fit Thread
ASME B16.5	-	Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard
ASME B18.21.1	-	Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)
ASME B46.1	-	Surface Texture (Surface Roughness, Waviness, and Lay)
ASME Boiler and Pressure Vessel Code Section III	-	Rules for Construction of Nuclear Facility Components

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ASME Boiler and Pressure Vessel Code Section VIII	-	Rules for Construction of Pressure Vessels Division 1

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A36/A36M	-	Standard Specification for Carbon Structural Steel
ASTM A53/A53M	-	Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A105/A105M	-	Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A106/A106M	-	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A181/A181M	-	Standard Specification for Carbon Steel Forgings, for General-Purpose Piping
ASTM A285/A285M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength
ASTM A515/A515M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service
ASTM A516/A516M	-	Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
ASTM B43	-	Standard Specification for Seamless Red Brass Pipe, Standard Sizes
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B98/B98M	-	Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes
ASTM B122/B122M	-	Standard Specification for Copper-Nickel-Tin Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver), and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar
ASTM B139/B139M	-	Standard Specification for Phosphor Bronze Rod, Bar, and Shapes
ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B150/B150M	-	Standard Specification for Aluminum Bronze Rod, Bar, and Shapes
ASTM B151/B151M	-	Standard Specification for Copper-Nickel-Zinc Alloy (Nickel Silver) and Copper-Nickel Rod and Bar
ASTM B152/B152M	-	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM B169/B169M	-	Standard Specification for Aluminum Bronze Sheet, Strip, and Rolled Bar
ASTM B171/B171M	-	Standard Specification for Copper-Alloy Plate and Sheet for Pressure Vessels, Condensers, and Heat Exchangers
ASTM B209/B209M	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B241/B241M	-	Standard Specification for Aluminum and Aluminum-Alloy Seamless Pipe and Seamless Extruded Tube
ASTM B265	-	Standard Specification for Titanium and Titanium Alloy Strip, Sheet, and Plate
ASTM B271/B271M	-	Standard Specification for Copper-Base Alloy Centrifugal Castings

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ASTM B338	-	Standard Specification for Seamless and Welded Titanium and Titanium Alloy Tubes for Condensers and Heat Exchangers
ASTM B363	-	Standard Specification for Seamless and Welded Unalloyed Titanium and Titanium Alloy Welding Fittings
ASTM B369/B369M	-	Standard Specification for Copper-Nickel Alloy Castings
ASTM B381	-	Standard Specification for Titanium and Titanium Alloy Forgings
ASTM B395/B395M	-	Standard Specification for U-Bend Seamless Copper and Copper Alloy Heat Exchanger and Condenser Tubes
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM D5363	-	Standard Specification for Anaerobic Single-Component Adhesives (AN)
ASTM F467	-	Standard Specification for Nonferrous Nuts for General Use
ASTM F468	-	Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AMS-P-83461	-	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275°F (135°C)
SAE AMS7259	-	Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant Low Compression Set/85 to 95 Hardness For Seals In Fuel Systems and Specific Engine Oil Systems
SAE AMS7276	-	Rubber: Fluorocarbon (FKM) High Temperature - Fluid Resistant Low Compression Set/70 to 80 Type A Hardness For Seals in Fuel Systems and Specific Engine Oil Systems
SAE J1926-1	-	Connections for General Use and Fluid Power - Ports and Stud Ends with ASME B1.1 Threads and O-Ring Sealing - Part 1: Threaded Port with O-Ring Seal in Truncated Housing

(Copies of these documents are available online at www.sae.org.)

TUBULAR EXCHANGER MANUFACTURERS ASSOCIATION (TEMA)

TEMA Standards	-	Class C - Commercial and General Process Applications

(Copies of this document are available online at www.tema.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 General requirements.

3.2.1 Design conditions. Cooler design conditions shall be as follows:

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a. For the fluid to be cooled, fluid type, design and peak flow rates, inlet and outlet temperatures, design pressure (see 6.8.3), and required rate of heat transfer shall be as specified (see 6.2).

b. For the coolant fluid, fluid type, design flow rate, inlet and outlet temperatures, and design pressure shall be as specified (see 6.2).

c. To ensure cooler design conditions are not overly constrained, inlet and outlet temperatures may be deferred to the equipment manufacturer by the procuring authority to meet the required rate of heat transfer. In such cases, applicable inlet or outlet temperatures shall be specified as "Vendor to Determine."

3.2.2 Space and weight. The occupied space and weight of coolers shall be held to a minimum consistent with meeting the requirements of this specification under the specified design conditions (see 6.2).

3.2.3 Ship attitude and motion. Unless otherwise specified (see 6.2), coolers shall meet the requirements in accordance with DOD-STD-1399-301 for ship motion and attitude.

3.2.4 Ambient pressure. For submarine application, coolers shall operate at an ambient pressure of 30 inches of mercury absolute with a variation of ±6 inches of mercury and shall not be damaged when subjected to an ambient pressure of 10 to 30 pounds per square inch (lb/in^2), absolute, with the internal pressure held to the minimum that will prevail under any condition of operation.

3.2.5 Prohibited materials. The following materials shall not be used for service, manufacture, test, or inspection of coolers covered in this specification:

a. Chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

b. Mercury and mercury compounds.

c. Beryllium and beryllium compounds.

d. Cadmium and cadmium-based alloys.

e. Magnesium and magnesium-based alloys.

3.2.6 Welding and allied processes.

3.2.6.1 Fabrication, welding, and inspection. Fabrication, welding, and inspection shall be in accordance with S9074-AR-GIB-010/278, except as modified herein.

3.2.6.2 Type I, class 1 coolers. For type I, class 1 coolers, all welding on parts subjected to sea pressure, including the entire double-tubesheet assembly, shall be in accordance with class A-F of S9074-AR-GIB-010/278, except as modified herein. Welded joints on the seawater side shall be radiographable except for vent and drain nipples. Waterbox vent and drain nipple welded connections shall meet the 100 percent weld-efficiency requirement for pressure vessels. Welded joints used in shell construction shall be in accordance with class A-4 of S9074-AR-GIB-010/278.

3.2.6.3 Class A-4 pressure vessels. For coolers classified as class A-4 pressure vessels in accordance with S9074-AR-GIB-010/278, weld joints P-64 and P-66 in accordance with MIL-STD-22 may be used for waterbox and shell nozzle attachments.

3.2.6.4 Weld root crevices. All weld root crevices exposed to lubricating oil shall be removed to prevent the trapping of contaminants.

3.2.6.5 Brazing. Brazing shall be in accordance with 0900-LP-001-7000, except that requirements for use of preinserted brazing rings are applicable only to pipe fittings. Brazed joints shall permit ultrasonic inspection. Brazing shall not be used for the following applications:

a. Joints for which there is a cyclic life requirement (for example, the seawater side of a type I, class 1 cooler).

b. Shells of coolers for submarine service.

c. Shell-pipe-to-shell-hub joints of nominal 10-inch diameter and larger.

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3.2.7 Casting inspection and repair. Casting inspection and repair shall be in accordance with S9074-AR-GIB-010/278. If cast construction is used for waterboxes of type I, class 1 coolers, they shall be in accordance with S9074-AR-GIB-010/278 for castings of category 2, subcategory G, except that no substitution of test shall be made. The extent of radiography shall be 100 percent to level 1 acceptance criteria, and the acceptance criteria for liquid penetrant inspection shall be class 1.

3.2.8 Threaded fasteners. Threaded fasteners shall be in accordance with MIL-DTL-1222, except that materials shall be as specified in 3.3.2 and 3.4.1. Configuration shall be such that standard wrenches can be used throughout. Tapered pipe threads shall not be used. The practice of "bottoming" or "shouldering" studs shall not be used (see 6.8.1 and 6.8.5). For the set end of studs, a class 3 fit in accordance with ASME B1.1 used with a locking compound in accordance with grade AN-0123 or AN-0124 of ASTM D5363 or a class 5 interference fit in accordance with ASME B1.12 shall be used.

3.2.8.1 Screw threads. Thread form, dimensions, and tolerances shall be in accordance with FED-STD-H28.

3.2.8.2 Thread lubricants. Threaded fasteners that have torque requirements shall be lubricated before assembly. The lubricant shall meet the following requirements, as applicable:

a. For propulsion plant coolers on saturated steam plant ships, the lubricant shall be in accordance with MIL-L-24478 or A-A-59004.

b. For all other bolting, the lubricant shall be in accordance with A-A-59004.

3.2.8.3 Preferred fastener types. Preferred threaded fastener types shall be as follows, in order of preference:

a. Through bolt or through (two-nut) stud.

b. Tap-end stud (one nut).

c. Cap screw.

3.2.8.4 Cap screws. Cap screws shall not be used for waterbox-to-shell (or waterbox-to-tubesheet) bolting, for inspection cover bolting, or for corrosion-preventive anode support cover bolting.

3.2.8.5 Waterbox and tubesheet bolting. For all waterbox and tubesheet bolting, the following requirements shall apply:

a. Bolting shall be tightened using a torque wrench to the torque values and in the order determined by the equipment manufacturer to obtain the necessary clamping force for the application.

b. For type I, class 1 coolers, bolting design, including stipulated torque values and supporting calculations, shall be established.

c. For type I, class 1 coolers, bolting subject to submergence pressure shall meet the requirements specified in 3.3.15.3.

3.2.8.6 Wrench flats. Where collar bolts or stud bolts are used to make up a multiple-flange joint so that it can be selectively disassembled, the bolts shall have a square extension (or pair of opposed flats) beyond the threads on one end for use of a wrench to prevent turning when the nuts are tightened or removed.

3.2.8.7 Internal fasteners. Where nuts are used inside the shell, tack-welding shall be used to prevent the nuts from coming adrift. For the set end of tie rods, a class 3 fit in accordance with ASME B1.1 used with a locking compound in accordance with grade AN-0123 or AN-0124 of ASTM D5363 or a class 5 interference fit in accordance with ASME B1.12 shall be used.

3.2.9 Shock resistance. The cooler shall pass the shock test in accordance with MIL-DTL-901 and as specified in 4.2.3. Shock test extension criteria in accordance with MIL-DTL-901 shall be adhered to with the following exception: a shock test extension request shall not be granted if the request is for a cooler that is longer or larger in diameter than the previously approved cooler.

3.2.10 Allowable pressure drop. Unless otherwise specified (see 6.2), the fluid side pressure drops (lb/in^2) under design conditions shall not exceed the values in table I when tested in accordance with 4.2.2.

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TABLE I. Allowable pressure drops.

Cooler type	Fluid side	Allowable pressure drop (lb/in^2)
Main propulsion turbine lubricating oil coolers that use scoop injection of circulating water	Tube (seawater) side	4
	Shell (oil) side	15
Other lubricating oil coolers and freshwater coolers	Coolant side	6
	Cooler fluid side (oil)	12
	Cooled fluid side (freshwater)	6
Hydraulic oil coolers	Tube (coolant) side	6
	Shell (oil) side	25

3.2.11 Cooling water velocity limits. Cooling water velocities shall not exceed those shown in table II. For coolers using seawater as the coolant, velocity through tubes shall be a minimum of 4.5 feet per second. (In determining shell-side velocities, class C TEMA Standards should be taken into consideration, but the shell-side velocities shall not exceed those shown in table II for velocity through tubes.)

TABLE II. Maximum cooling water velocities.

Coolant and method of supply	Velocity through inlet flange (feet per second)	Velocity through tubes (feet per second)
Seawater, supplied by scoop injection	11.0	9.0
Seawater, otherwise supplied	9.0	7.5
Fresh water, however supplied	11.0	9.0

3.2.12 Cooling water inlet temperature. Unless otherwise specified (see 6.2), for design purposes, temperatures of coolant supply shall be taken as specified in table III.

TABLE III. Cooling water inlet temperature.

Ship or craft	Coolant	Temperature (°F)
Submarines	Seawater	85
	Fresh water	95
Large surface ships	Seawater (electronic system)	95
	Seawater (other systems)	90
	Fresh water	95
Patrol boats and small craft	Seawater	95
	Fresh water	100

3.2.13 Heat transfer surface. The amount of heat transfer surface installed in coolers shall be based on the following:

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a. For submarine coolers, a 10 percent fouling factor shall be applied to the service heat transfer coefficient for clean tubes.

b. For surface ship coolers, the service heat transfer coefficient shall be calculated in accordance with the fouling resistances given in table IV. For fluid not listed in table IV, the fouling resistance in accordance with class C TEMA Standards shall be used.

c. If the cooler is designed to use finned tubing, an appropriate correction factor shall be used in the heat transfer analysis, accounting for fin geometry. The correction factor shall be used to correct the predicted shell-side heat transfer coefficient to account for the laminar boundary layer overlap on the fins.

TABLE IV. Fouling resistances (for calculating service heat transfer coefficients for surface ship coolers).

Fluid		Resistance (R), (hr ft^2 °F)/Btu
Seawater		0.0005
Fresh water		0.0005
Lubricating oil		0.001
Hydraulic oil		0.001
Fuel oil (R to vary with viscosity)	100 Saybolt universal second (SSU) (at 100 °F) or less	0.002
	100 to 500 SSU (at 100 °F)	0.003
	500 to 2000 SSU (at 100 °F)	0.004
	2000 SSU (at 100 °F) or greater	0.005

3.2.14 Inlet and outlet connections. Inlet and outlet connections for coolant and for cooled fluid shall be of the size specified and shall be flanged unless hose connections are specified (see 6.2). Connection flanges shall be of the cast-integral or the welding type and are further described in 3.2.15.

3.2.15 Connection flanges. Unless otherwise specified (see 6.2), requirements for flanged connections shall be as follows:

a. For surface ships, flanges shall be in accordance with MIL-STD-777. For submarines, MIL-STD-438 or equivalent (when specified [see 6.2]) should be used as a guide in regard to diameter, minimum thickness, drilling, and flange finish, except as specified herein.

b. For flanges using O-ring gaskets, the groove for the O-ring shall be in the companion flange on the shipbuilder's piping.

c. For nonferrous applications, the requirements for flanges specified in MIL-PRF-20042 shall constitute minimum standards.

d. For ferrous applications, the 150-pound primary service pressure rating requirements for flanges in accordance with ASME B16.5 shall constitute minimum standards.

e. For cast waterboxes, shells, shell hubs, and the equivalent, cast-integral nozzle flanges may be used in cases where connections are expressed in terms of pipe-flange standards; for these cases, requirements for flange identification markings shall be waived.

3.2.15.1 Surface finish of gasket mating surfaces. The machine surface finish of gasket mating surfaces on flanges in piping systems and connected components shall be in accordance with ASME B46.1 and as follows:

a. Nonferrous and ferrous mating surfaces for use with sheet gaskets:

(1) For flanges of a nominal pipe size (NPS) of 12 inches or less, a finish with a circular lay of 500 to 1000 (concentric) or 125 to 250 (spiral) microinches roughness average (RA) produced by machining 30 to 80 serrations of uniform depth per inch of face width.

(2) For flanges over an NPS of 12 inches, the requirements shall be the same as above except that 21 to 80 serrations per inch of face width may be used.

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(3) For flanges where the flange face cannot be turned and tool marks run across the flange face, the surface finish shall be 63 to 125 microinches RA.

b. Flanges for O-ring seals:

(1) A finish of 32 microinches RA maximum in O-ring grooves.

(2) A finish of 63 or smoother microinches RA on the flange face opposite the groove.

3.2.16 Fill, vent, and drain connections. Coolant and cooled-fluid circuits shall be constructed in such a way to permit complete venting and draining. Coolers with seawater as the coolant shall have attached fill, drain, and vent connections necessary to support freshwater lay-up of the seawater side. If double-tubesheet construction is specified, the inter-tubesheet spaces shall have vent and drain connections. Vent and drain lines shall be not less than ¼-inch NPS. Installer-furnished vent and drain tubing to be connected to the cooler shall be of the size, wall thickness, and material specified (see 6.2). Root connections shall have wall thickness in accordance with MIL-T-16420; 70-30 copper-nickel tube (class 6000 for 1-inch NPS and smaller; class 3300 for 1.050-inch NPS and larger) and shall be as follows:

a. For coolant-side and inter-tubesheet spaces of submarine cooler applications designed for seawater cooling, full penetration weld with joint design shall be in accordance with P-70, P-71, or P-72 of MIL-STD-22. Nipples shall be composition 70-30 copper-nickel in accordance with MIL-C-15726.

b. For other applications, in addition to the method specified in 3.2.16.a, the following methods may be used:

(1) Single-J, fillet-reinforced weld V-27 in accordance with MIL-STD-22. Nipples shall be copper-nickel in accordance with MIL-T-16420.

(2) Tapped hole in accordance with SAE J1926-1 with an adapter, except fitting shall be composition 90-10, copper-nickel alloy, in accordance with MIL-C-15726. The adapter and threads on the cooler shall be sized to withstand, without yielding, loads sufficient to cause yielding of the installer-furnished vent and drain tubing.

c. In addition to methods specified in 3.2.16.b, for applications where the connection will not normally be in contact with seawater, a socket weld using a T by 2T minimum fillet in accordance with P-14 of MIL-STD-22 may be used.

3.2.16.1 High-pressure cooler vent and drain egress. For high-pressure applications, waterbox vents and drain connections shall be led out by means of the waterbox flange or tubesheet.

3.2.17 Pressure gauge connections. When specified (see 6.2), connections for pressure gauges shall be provided at inlets and outlets of coolant and cooled-fluid sides. Gauge connections shall conform to 3.2.16.

3.2.18 Supports. Coolers shall be provided with supports for securing to a foundation or parent machine as specified (see 6.2). The means of support shall be independent of connecting piping. For coolers designed to be supported from a vertical structure such as a bulkhead, the supports shall be such that bolts in shear will not constitute the primary means of support. Coolers shall not be supported by plates or brackets in such a manner that the primary means of support is obtained from the bolts that secure the joint between shell-end flange, tubesheet, and waterbox flange.

3.2.19 Foundation bolt hole allowances. The drilling of foundation bolt holes in cooler feet, other than for close-allowance bushing application (see 3.2.20), shall meet the following criteria:

a. For bolt sizes of ¾-inch diameter and smaller, the bolt hole shall be not larger than bolt diameter plus 1/32 inch.

b. For bolt sizes larger than ¾-inch diameter, the bolt hole shall be not larger than bolt diameter plus 1/16 inch.

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3.2.20 Shell expansion relative to supports. If the distance between cooler supports exceeds 24 inches, the shell shall be constructed in such a way that it expands and contracts with respect to its supports. The use of elongated or oversized bolt holes in support feet is not, by itself, an acceptable method. The following methods shall be used:

a. Providing freedom for one end of the shell to move in the axial direction by means of elongated holes in the foot of that shell support, with shell end motion in the other two planes being restricted by use of close-allowance collared bushing (collar may be integral or separate from the bushing), the shell shall be assembled over and rigidly bolted down by the foundation bolts at that end. When this method of support is used, the other (fixed) support and its foundation bolts shall carry the entire load due to high-impact shock in the direction longitudinal of the shell.

b. Providing freedom for one end of the shell to move in the axial direction by use of a support designed to flex as the shell expands or contracts at that end of the cooler. When this method of support is used, the other (fixed) support and its foundation bolts shall carry the entire load due to high-impact shock in the direction longitudinal of the shell. Where coolers have cast shells or shell hubs, the flexible support may be a separate piece bolted to the cast shell end.

3.2.21 Provision for shell expansion relative to tubes. For fixed bundle coolers (stationary tubesheets at both ends), a shell expansion joint shall be provided to compensate for differential expansion between shell and tubes when the tube bundle length (back-to-back of tubesheets) is greater than 18 inches. The expansion joint shall be located between one support and the shell-end flange.

3.2.22 Gasket application. Unless otherwise specified (see 6.2), flat gaskets may be used in lieu of O-ring gaskets for liquid-side applications of less than 200 lb/in^2 design pressure. O-ring gaskets shall be used for design pressures greater than 200 lb/in^2 gauge. Blind gaskets or packings (those not replaceable without destructive disassembly) shall not be used. Fluorocarbon rubber O-rings or gaskets shall not be used on the freshwater side of submarine coolers.

3.2.23 Flat gaskets. Unless otherwise specified (see 6.2), flat gaskets shall be 1/16 inch thick. Gasket width under waterbox partitions shall be not less than 5/16 inch. Flat gaskets shall conform to the material requirements of 3.3.2.1 through 3.3.2.5 with the following exceptions:

a. For type I cooler applications where design pressure does not exceed 100 lb/in^2, synthetic rubber in accordance with MIL-PRF-1149, type II, class 5 may be used.

b. Unless otherwise specified (see 6.2), for type I cooler applications where temperature of metal in contact with the gasket will exceed 250 °F, gasket material shall be in accordance with MIL-DTL-24696.

3.2.24 O-ring gaskets. O-ring gaskets for application in fresh water, seawater, or standard lubricating and petroleum hydraulic oil shall be in accordance with SAE AMS7276 (SAE AMS7259 for threaded adapters and plugs), except O-rings on the freshwater side of submarine coolers, which shall be in accordance with SAE AMS-P-83461, MIL-G-21610 type I, or as otherwise specified (see 6.2). O-ring segments for sealing under waterbox partitions shall be integral with the peripheral O-ring.

3.2.24.1 Gasket life. Gasketed closures that are not normally opened for preventive maintenance shall be designed to function under the specified design conditions of flow, temperature, and pressure (see 3.2.1) for a minimum life of 30 years. Gasketed closures that are normally opened for preventive maintenance shall be designed to function under the specified design conditions (see 3.2.1) for a minimum life of 10 years.

3.2.25 Selection of rubber packing materials. Rubber packing materials shall be selected to meet the temperature demands of the application and be compatible with the working fluids. Packing rings in accordance with class 5 of MIL-PRF-1149 shall be used with petroleum-based lubricating oils.

3.2.26 Identification plates. Each cooler shall bear an identification plate in accordance with class A, B, C, or D of MIL-DTL-15024 and MIL-P-15024/5, with material choice being limited to wrought brass, cast brass, cast bronze, or corrosion-resistant steel. (Plates shall be normal service plates.) Design pressure entries shall not appear on identification plates. The identification plates shall include the following information:

a. Name of unit (examples: lubricating oil cooler, freshwater cooler, battery cooler, hydraulic oil cooler).

b. Type of unit (see 1.2).

c. Class of unit (see 1.2).

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d. Manufacturer's service part number.

e. National stock number. Allow 20 spaces.

f. Name of manufacturer.

g. Contract or purchase order number. (Utilize width of plate to allow maximum number of spaces.)

h. Blank space for Defense Contract Management Agency (DCMA) stamp.

i. Date of manufacture.

j. Serial number.

k. Maximum factory test pressure, shell-side.

l. Maximum factory test pressure, tube-side. For coolers for seawater service on submarines, no entry shall be made in this space.

m. Blank space for unit number. When specified (see 6.2), a blank entry of four spaces shall be provided. This space will be used by the shipyard to stamp unit numbering for shipboard reference purposes.

n. Designation "U.S." (without quotation marks).

3.2.27 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength.

3.3 Type I (shell and tube design) requirements.

3.3.1 Code requirements for type I coolers. Type I coolers shall be constructed in accordance with class C TEMA Standards, except as otherwise specified herein. The structural design calculations for the class 1 coolers shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section III, class 1 vessel. Structural design calculations for coolers other than class 1 shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section VIII, Division I.

3.3.2 Materials. For type I coolers, the materials specified below shall be used as the basis for design. These shall be used in the construction of the coolers.

3.3.2.1 Type I, class 1. Type I, class 1 coolers shall be constructed of the materials specified in table V.

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TABLE V. Type I, class 1 cooler materials.

Part	Material	Applicable document
Shells 1/	Copper-nickel alloy	MIL-C-15726
	Tubing, copper-nickel alloy	MIL-T-16420
Waterboxes 2/, 6/	Copper-nickel alloy, composition 70-30	MIL-C-15726
	Tubing, copper-nickel alloy, composition 70-30	MIL-T-16420
	Copper-nickel alloy, C96400 70-30	ASTM B369/B369M
	Bronze, nickel-aluminum casting	MIL-B-24480
Tubesheets 3/, 6/	Copper-nickel alloy, composition 70-30	MIL-C-15726
Tubes 6/	Copper-nickel alloy, composition 70-30	MIL-T-15005
Gland rings and lantern rings	Bronze, tin, centrifugal castings, alloy C90300 or C92200	ASTM B271/B271M
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
	Copper-nickel alloy	MIL-C-15726
Baffles	Copper-nickel alloy	MIL-C-15726
Spacers	Copper-nickel alloy	MIL-T-15005
		MIL-T-16420
Spacer rods and spacer nuts	Copper-nickel alloy	MIL-C-15726
Bolts or studs in joints involving seawater tightness or subject to submergence pressure 4/, 6/	Nickel-copper-aluminum alloy, grade A, hot finished, annealed, and age hardened; or cold drawn, annealed, and age hardened	MIL-DTL-1222
Nuts for bolts and studs in joints involving seawater tightness or subject to submergence pressure 4/, 6/	Nickel-copper-aluminum alloy	MIL-DTL-1222, grade 500
	Nickel-copper alloy	MIL-DTL-1222, grade 400
	Nickel-copper alloy, cold drawn and stress relieved, or hot finished	MIL-DTL-24106
Bolts, studs, and nuts for service other than specified above 4/	Nickel alloys, grade 400 or 405	MIL-DTL-1222
Jackscrews	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
	Phosphor bronze	ASTM B139/B139M
	Copper-silicon alloy	ASTM B98/B98M
Corrosion preventive anodes	Zinc	MIL-DTL-18001
Flat gaskets 5/	Rubber, synthetic, class 1, grade 80	MIL-PRF-6855
		MIL-DTL-24696
Washers	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
		ASTM B169/B169M
		QQ-C-450
Plugs and adapters	Copper-nickel alloy, composition 70-30	MIL-C-15726
Packing rings	Rubber, synthetic, type II, class 5	MIL-PRF-1149

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TABLE V. Type I, class 1 cooler materials – Continued.

Part	Material	Applicable document
O-ring gaskets [5/]	Rubber, synthetic, type I	MIL-G-21610
	Rubber, synthetic	SAE AMS7276
		SAE AMS7259
		SAE AMS-P-83461

NOTES:

[1/] Steel may be used for shell support members.

[2/] Inspection covers shall be of the same copper alloy as the waterboxes to which they are fitted.

[3/] Forged or wrought material shall be ultrasonically inspected in accordance with MIL-C-15726.

[4/] Material for studs, bolts, and nuts shall be ultrasonically inspected in accordance with MIL-DTL-1222.

[5/] See 3.2.22, 3.2.23, and 3.2.24.

[6/] Designated as critical material as these components are seawater-connected systems that are subject to submergence pressure below 200 feet. Each of these items shall have material identification and control as specified (see 6.2).

3.3.2.2 Type I, class 2. Type I, class 2 coolers shall be constructed of the materials specified in table VI.

TABLE VI. Type I, class 2, cooler materials.

Part	Material	Applicable document
Shells [1/]	Copper-nickel alloy	MIL-C-15726
	Tubing, copper-nickel alloy	MIL-T-16420
Waterboxes [2/]	Copper-nickel alloy, composition 70-30	MIL-C-15726
	Tubing, copper-nickel alloy, composition 70-30	MIL-T-16420
	Copper-nickel alloy, C96400 70-30	ASTM B369/B369M
	Bronze, nickel-aluminum casting	MIL-B-24480
Tubesheets [3/]	Copper-nickel alloy, composition 70-30	MIL-C-15726
Tubes	Copper-nickel alloy, composition 70-30	MIL-T-15005
Gland rings and lantern rings	Bronze, tin, centrifugal castings, alloy C90300 or C92200	ASTM B271/B271M
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
	Copper-nickel alloy	MIL-C-15726
Baffles	Copper-nickel alloy	MIL-C-15726
Spacers	Copper-nickel alloy	MIL-T-15005
		MIL-T-16420
Spacer rods and nuts	Copper-nickel alloy	MIL-C-15726

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TABLE VI. Type I, class 2, cooler materials – Continued.

Part	Material	Applicable document
Bolts or studs in joints involving seawater tightness	Nickel alloy, grade 400, 405, or 500	MIL-DTL-1222
Nuts for bolts and studs in joints involving seawater tightness	Nickel alloy, grade 400, 405, or 500	MIL-DTL-1222
Bolts, studs, and nuts for service other than specified above	Nickel alloys	MIL-DTL-1222
Jackscrews	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
	Phosphor bronze, C51000, C54400, or C52400	ASTM B139/B139M
	Copper-silicon alloy C66100	ASTM B98/B98M
Corrosion preventive anodes	Zinc	MIL-DTL-18001
Flat gaskets 4/	Rubber, synthetic, class 1, grade 80	MIL-PRF-6855
		MIL-DTL-24696
	Rubber, synthetic, type II, class 5	MIL-PRF-1149
Washers	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
		ASTM B169/B169M
Plugs and adapters	Copper-nickel alloy, composition 70-30	MIL-C-15726
Packing rings	Rubber, synthetic, type II, class 5	MIL-PRF-1149
O-ring gaskets 4/	Rubber, synthetic, type I	MIL-G-21610
	Rubber, synthetic	SAE AMS7276
		SAE AMS7259
		SAE AMS-P-83461

NOTES:

1/ Steel may be used for shell support members.

2/ Inspection covers shall be of the same copper alloy as the waterboxes to which they are fitted.

3/ For double-tubesheet construction, an additional material for inner tubesheets is copper-nickel alloy, composition 90-10, in accordance with MIL-C-15726.

4/ See 3.2.22, 3.2.23, and 3.2.24.

3.3.2.3 Type I, class 3. Type I, class 3 coolers shall be constructed of the materials specified in table VII.

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TABLE VII. Type I, class 3 cooler materials.

Part	Material	Applicable document
Shells [1/]	Copper-nickel alloy, composition 90-10	MIL-C-15726
	Tubing, copper-nickel alloy, composition 90-10	MIL-T-16420
	Copper-nickel alloy	ASTM B171/B171M
	Copper-nickel alloy, alloy C70600 or C71500, temper 0S035	ASTM B151/B151M
	Copper-aluminum alloy, copper alloy C61400	QQ-C-450
	Bronze, tin, sand castings, alloy C90500 or C92200	ASTM B584
	Bronze, nickel-aluminum casting	MIL-B-24480
Waterboxes [2/]	Copper-nickel alloy	MIL-C-15726
	Tubing, copper-nickel alloy	MIL-T-16420
	Copper-nickel alloy	ASTM B369/B369M
	Bronze, nickel-aluminum casting	MIL-B-24480
	Bronze, sand castings, alloy C90500 or C92200	ASTM B584
Tubesheets [3/]	Copper-nickel alloy, composition 90-10	MIL-C-15726
	Bronze, nickel-aluminum, alloy C63000	ASTM B171/B171M
Tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
		MIL-T-22214
Outer tubes, for double-tube construction	Copper	MIL-T-24107
Fins	Nonferrous	----
Gland rings and lantern rings	Bronze, tin, centrifugal casting alloy C90300 or C92200	ASTM B271/B271M
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
	Copper-nickel alloy	MIL-C-15726
Baffles	Copper-nickel alloy	MIL-C-15726
Spacers	Copper-nickel alloy	MIL-T-15005
		MIL-T-16420
Spacer rods and spacer rod nuts	Copper-nickel alloy	MIL-C-15726
Bolts, studs, and nuts	Nickel alloys	MIL-DTL-1222
Jackscrews	Copper alloys	MIL-DTL-1222
Corrosion preventive anodes	Zinc	MIL-DTL-18001
Plugs, zinc support	Copper-nickel alloy, composition 70-30	MIL-C-15726
Flat gaskets [4/]	Rubber, synthetic, class 1, grade 80	MIL-PRF-6855
		MIL-DTL-24696
	Rubber, synthetic, type II, class 5	MIL-PRF-1149
Washers	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
		ASTM B169/B169M

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TABLE VII. Type I, class 3 cooler materials – Continued.

Part		Material	Applicable document
Plugs and adapters	For all services	Nickel-copper alloy	QQ-N-281
		Copper-nickel alloys	MIL-C-15726
		Bronze tin, sand castings, alloy C90300 or C92200	ASTM B584
		Copper-nickel alloy, alloy C71500, temper 0S035	ASTM B122/B122M
	For other than seawater side	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
		Nickel-aluminum bronze, alloy C63200	QQ-C-450
			ASTM B150/B150M
Packing rings		Rubber, synthetic, type II, class 5	MIL-PRF-1149
O-ring gaskets [4/]		Rubber, synthetic, type I	MIL-G-21610
		Rubber synthetic	SAE AMS7276
			SAE AMS7259
			SAE AMS-P-83461

NOTES:

[1/] Steel may be used for shell support members.

[2/] Inspection covers shall be of the same copper alloy as the waterboxes to which they are fitted.

[3/] For double-tubesheet construction, additional materials for inner tubesheets shall be copper-aluminum alloys, copper alloys C61300 and C61400 in accordance with QQ-C-450, and copper-nickel alloy, composition 90-10, in accordance with MIL-C-15726.

[4/] See 3.2.22, 3.2.23, and 3.2.24.

3.3.2.4 Type I, class 4. Unless otherwise specified (see 6.2), type I, class 4 coolers shall be constructed of the materials specified in table VIII.

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TABLE VIII. Type I, class 4 and 5 cooler materials.

Part	Material	Applicable document
Shells [1/]	Copper-nickel alloy	MIL-C-15726
	Tubing, copper-nickel alloy	MIL-T-16420
	Copper-nickel alloy, alloy C70600 or C71500	ASTM B171/B171M
	Copper-nickel alloy, alloy C70600 or C71500, temper 0S035	ASTM B151/B151M
	Copper-aluminum alloy, copper alloy C61400	QQ-C-450
	Bronze, tin, sand castings, alloy C90500 or C92200	ASTM B584
	Bronze, nickel-aluminum casting	MIL-B-24480
Waterboxes	Copper-nickel alloy	MIL-C-15726
	Tubing, copper-nickel alloy	MIL-T-16420
	Copper-nickel alloy	ASTM B369/B369M
	Bronze, nickel-aluminum casting	MIL-B-24480
	Aluminum bronze, alloy C95400 as cast, or alloy C95200	ASTM B148
	Bronze, tin, sand castings, alloy C90500 or C92200	ASTM B584
Tubesheets [2/]	Copper-nickel alloy	MIL-C-15726
	Nickel-aluminum, bronze, alloy C63000	ASTM B171/B171M
	Aluminum bronze, nickel-aluminum, alloy C61400	ASTM B171/B171M
Tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
		MIL-T-22214
Gland rings and lantern rings	Bronze, tin, centrifugal castings, alloy C90300 or C92200	ASTM B271/B271M
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
	Copper-nickel alloy	MIL-C-15726
Baffles	Copper-nickel alloy	MIL-C-15726
Spacers	Copper-nickel alloy	MIL-T-15005
		MIL-T-16420
Spacer rods and spacer rod nuts	Copper-nickel alloy	MIL-C-15726
Bolts, studs, and nuts	Nickel-copper, grade 400 or 405	MIL-DTL-1222
	Nickel-copper-aluminum, grade 500	
Jackscrews	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
	Phosphor bronze C51000, C54000, or C52400	ASTM B139/B139M
	Copper-silicon alloy C66100	ASTM B98/B98M
Flat gaskets [3/]	Rubber, synthetic, class 1, grade 80	MIL-PRF-6855
	Rubber, synthetic, type II, class 5	MIL-PRF-1149
Washer	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
		ASTM B169/B169M
		QQ-C-450

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TABLE VIII. Type I, class 4 and 5 cooler materials – Continued.

Part	Material	Applicable document
Plugs and adapters	Copper-nickel alloy	MIL-C-15726
	Copper-nickel alloy, alloy C71500, temper 0S035	ASTM B122/B122M
Packing rings	Rubber, synthetic, type II, class 5	MIL-PRF-1149
O-ring gaskets 3/	Rubber, synthetic, type I	MIL-G-21610
	Rubber synthetic	SAE AMS7276
		SAE AMS7259
		SAE AMS-P-83461

NOTES:

1/ Steel may be used for shell support members.

2/ Material for type I, class 4 cooler tubesheets shall be copper-nickel alloy in accordance with MIL-C-15726. Material shall be 100 percent ultrasonically inspected in accordance with MIL-C-15726.

3/ See 3.2.22, 3.2.23, and 3.2.24.

3.3.2.5 Type I, class 5. Unless otherwise specified (see 6.2), type I, class 5 coolers shall be constructed of the materials specified in table VIII or table IX.

TABLE IX. Alternative type I, class 5 cooler materials.

Part	Material	Applicable document
Waterboxes	Aluminum-bronze, alloy C95400 as cast, or alloy C95200	ASTM B148
Tubesheets	Copper-aluminum alloy, copper alloy C61300	QQ-C-450
	Aluminum bronze D, copper alloy C61400	ASTM B171/B171M
Tubes	Copper-nickel alloy, composition 90-10	MIL-T-15005
		MIL-T-22214

3.3.2.6 Additional materials allowances. In addition to the materials listed in tables V through IX, the following material allowances apply to the construction of lubricating and hydraulic oil coolers.

3.3.2.6.1 Use of steel material for the shell side of lubricating oil and hydraulic oil coolers. Unless otherwise specified (see 6.2), for lubricating oil and hydraulic oil cooling applications, as subject to the material limitations in MIL-STD-2193 for hydraulic service, the shell side of coolers may be constructed of the steel materials specified in table X.

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TABLE X. Optional shell-side materials for oil cooling applications.

Part	Material	Applicable document
Shell	Steel pipe, type S	ASTM A53/A53M
	Seamless carbon steel pipe, grade B	ASTM A106/A106M
	Steel plate, grade C	ASTM A285/A285M
	Seamless drawn steel tubing	MIL-P-24691/1
	Steel plate	MIL-S-22698
	Cast steel	MIL-DTL-24707/1
	Carbon steel plate	ASTM A515/A515M
		ASTM A516/A516M
Flanges	Carbon steel plate	ASTM A515/A515M
		ASTM A516/A516M
	Steel plate, grade C	ASTM A285/A285M
	Cast steel	MIL-DTL-24707/1
	Steel pipe flange	ASTM A181/A181M
	Carbon steel forgings	ASTM A105/A105M
Shell supports	Carbon steel	ASTM A105/A105M
	Carbon steel plate	ASTM A36/A36M
		ASTM A515/A515M
		ASTM A516/A516M
	Cast steel, grade A1Q	ASTM A757
	Grade WCA	ASTM A216
		MIL-DTL-24707/1
	Steel plate, grade C	ASTM A285/A285M
	Steel plate, class C	ASTM A515/A515M
		ASTM A516/A516M
Tube, outer	Brass, class A	ASTM B43

3.3.2.6.2 Use of aluminum alloy shell-side internals for lubricating oil and hydraulic oil coolers. When specified (see 6.2), for lubricating oil and hydraulic oil cooling applications, the shell-side internal part of type I, class 1 coolers may be constructed of the aluminum alloys specified in table XI.

TABLE XI. Optional materials for shell-side internal parts for oil cooling applications, type I coolers.

Part	Material	Applicable document
Baffles	Aluminum, alloy 2024-T3	ASTM B209/B209M
Spacers	Aluminum, alloy 6061-T6	ASTM B241/B241M
Tie rods	Aluminum, alloy 6061-T6	ASTM F468
Tie rod nuts	Aluminum, alloy 6062-T9	ASTM F467

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3.3.2.6.3 Use of titanium materials for lubricating oil and hydraulic oil coolers. When specified (see 6.2), for lubricating oil and hydraulic oil cooling applications, type I, class 3 coolers may be constructed of the steel and titanium materials specified in table XII.

TABLE XII. Optional materials for type I, class 3 titanium tube coolers for lubricating and hydraulic oil cooling applications.

Part	Material	Applicable document
Waterboxes	Titanium, grade 2	ASTM B265
Waterbox nozzles	Titanium, grade 2	ASTM B265 or ASTM B381
Waterbox nozzle flanges	Titanium, grade 2 or grade 3	ASTM B363 or ASTM B265
Tubesheets	Carbon steel plate clad with Titanium, grade 2	ASTM A516/A516M, ASTM B265, or ASTM B381
Tubes	Titanium, grade 2	ASTM B338
Baffles	Carbon steel	ASTM A516/A516M
Bolts, studs, capscrews, and nuts	Nickel-copper alloy	QQ-N-281

3.3.3 Tube features. Cooler tubes shall be seamless. Coolers shall have straight tubes unless U-bend tubes are specified (see 6.2).

3.3.3.1 U-bend tubes. The bending method shall be approved for each combination of tube size, gauge, fin geometry, material, tube manufacturer, and tube-bending vendor proposed for use (see 6.2 and appendix A). Demonstration of approval shall be by destructive examinations of minimum bend radius samples for wall thinning and roundness in accordance with ASTM B395/B395M. U-bend tubes shall be nondestructively tested in accordance with ASTM B395/B395M. Bending radii for U-bend tubes shall be not less than the dimensions specified in table XIII.

TABLE XIII. Minimum radii for U-bend tubes, type I coolers.

Outside tube diameter (inch)	Minimum radius of bend at tube centerline (inch)
5/8	15/16
1/2	13/16
3/8	5/8

3.3.3.1.1 U-bend finned tubes. U-bend finned tubes shall be annealed after finning prior to bending.

3.3.4 Shell-end construction. Shell-end construction shall be as follows (see 3.3.5):

a. Ring flange, drilled for bolting of tubesheets and bonnet (as applicable) thereto, with shell welded to the flange or cast integral with it. The finished sealing surface of the flange shall meet a flatness tolerance of 0.020 inch and the surface finish requirements specified in 3.2.15.1.

b. Additionally, for cooling applications where double-tube construction is specified (see 3.3.8), shell flange omitted, shell welded to tubesheet.

3.3.5 Cooler end construction. Except for cooling applications where double-tube construction is specified (see 3.3.5.3 and 3.3.8), the cooler shall have a removable tube bundle with a stationary tubesheet at the front end of the cooler (see 6.8.4). The removable feature shall be achieved by use of U-bend-tube construction or by use of straight-tube construction in conjunction with a floating tubesheet at the rear end of the cooler. Standard construction for these coolers shall conform to 3.3.5.1 and 3.3.5.2, as applicable.

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3.3.5.1 Submarine cooler applications (except type I, class 4, applications not exceeding 150 lb/in^2 design pressure).

a. Front-end construction shall be either of the following:

(1) Gasket or O-ring joint for tubesheet-to-shell flange; gasket or O-ring joint for waterbox-to-tubesheet; tubesheet secured to shell-flange by use of collar studs or studs threaded into the tubesheet to ensure the tubesheet-to-shell flange joint withstands shell-side hydrostatic testing with the waterbox disassembled.

(2) Full face gasket or O-ring joint for tubesheet-to-shell flange with independent bolting ring; reduced diameter stationary tubesheet-to-waterbox flange gasket or O-ring joint secured by independent bolting ring.

b. Rear-end construction where a floating tubesheet is used shall be as follows (see figure 1): The waterbox shall be attached to the floating tubesheet by means of a ring of studs set in the face of the tubesheet and used to form a gasketed joint between the waterbox and the tubesheet. The seal between the shell and the floating tubesheet shall be effected by rubber or rubberlike packing rings. The packing rings shall fit in a counterbore in the shell end and be compressible by a gland ring held in place by nuts on studs set in the shell end. The perimeter of the floating tubesheet (or an extension of it) shall bear against the concave side of the packing rings in the manner of a piston in a cylinder.

3.3.5.2 Submarine cooler (type I, class 4, not exceeding 150 lb/in^2 design pressure) and surface ship cooler applications.

a. Front-end construction shall be in accordance with 3.3.5.1.a or as follows: Gasket or O-ring joint for the tubesheet-to-shell flange; gasket or O-ring joint for the waterbox-to-tubesheet joint; tubesheet secured to shell flange by use of collar studs or studs threaded into the tubesheet so that the tubesheet-to-shell flange joint can withstand shell-side hydrostatic testing with the waterbox disassembled. For U-bend-tube construction, all the fasteners of the bolting ring shall be of one of the types described above; for floating tubesheet construction, not less than 50 percent of them shall be of such construction.

b. Rear-end construction where a floating tubesheet is used shall be in accordance with 3.3.5.1.b or either of the following:

(1) The shell and waterbox flanges shall each be counterbored to retain one or more integral packing rings. A lantern ring shall be located between the respective packing rings or sets of packing rings. This lantern ring shall have a groove around the inside, and leakoff holes shall be arranged so that leakage past the packing ring (whether from the shell or the waterbox) will be relieved to the outside. The waterbox-to-shell bolting shall compress this double-packed joint. Where this packed lantern ring construction is used, a test ring shall be provided so that hydrostatic testing of the shell side can be accomplished with the rear waterbox removed. Rear end construction of this kind is shown on figure 2.

(2) The shell and waterbox flanges shall have a packing gland retaining ring to hold one or more integral packing rings in place. A lantern ring shall be located between the respective packing rings or sets of packing rings. This lantern ring shall have a groove around the inside and leakoff holes arranged so that leakage past the packing ring (whether from the shell or the waterbox) will be relieved to the outside. The waterbox-to-shell bolting shall compress this double-packing joint. Where this construction is used, no test ring shall be required, since the waterbox stud bolts shall be threaded into the lantern ring and the lantern ring shall, therefore, remain in place after the waterbox is removed. This rear-end construction is shown on figure 3.

c. When specified (see 6.2), units shall feature coolant flow paths led through the shell with coolant piping connections integral with the shell. The stationary tubesheet shall be of reduced diameter relative to shell and waterbox flanges, but with joint configuration such that both the waterbox and the shell shall limit lateral motion of the stationary tubesheet. A test ring shall be provided to allow testing of the shell-side with the stationary waterbox removed.

3.3.5.3 Cooling applications for which double-tube construction is specified. For these coolers, the method specified in 3.3.4.b shall be used to attach the shell to the tubesheets. These coolers shall be of the following construction:

a. Shells shall be of fabricated construction. Straight tubes shall be used.

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b. If the tube bundle effective length (back-to-back of tubesheets) exceeds 18 inches, a shell expansion joint shall be provided to compensate for differential expansion between the shell and tubes. This joint shall be located outboard of the shell supports and shall withstand lateral high-impact shock. Stuffing box-type joints shall not be used.

3.3.6 Hydraulic oil coolers. For hydraulic oil cooling applications, coolers shall incorporate the double-tubesheet feature of 3.3.7. Type I, class 4 coolers not exceeding 150 lb/in^2 design pressure shall incorporate a removable tube bundle with a stationary tubesheet for front-end construction as specified in 3.3.5.2.a. When double-tube construction is specified (see 3.3.8), hydraulic oil coolers shall be of the fixed-bundle, straight-tube type specified in 3.3.5.3.

3.3.7 Double-tubesheet construction (cyclic loading not specified). For hydraulic oil coolers and as specified for other coolers (see 6.2), two tubesheets shall be provided at each end of the tubes. A space between the two sheets of a pair shall be provided by use of a spacing ring or by machining one or both sheets. The joint between the sheets or the joints between sheets and ring shall be welded (see figures 4 and 5). The void space between tubesheets and ring shall be vented and drained to atmosphere. The tubes shall be expanded into each tubesheet as specified in 3.3.10.3. After expansion of the tubes, the specified tube-side hydrostatic test pressure shall be applied to the void space of each double tubesheet, and there shall be no leakage under this pressure. Double-tubesheet construction subject to cyclic loading shall be as specified in 3.3.15.

3.3.8 Double-tube construction. When specified (see 6.2), double-tube construction shall be combined with fixed-bundle construction and double-tubesheet construction. If this combination is used, a telltale system shall be used for detecting leaks. Outer tubes shall be jointed to the inner tubesheets by expanding, welding, or silver brazing. The outer tubes shall have their internal surfaces longitudinally grooved so that after the inner tubes are expanded tightly into the outer tubes, the contact between the two tubes conducts heat sufficient to meet the heat transfer requirements specified (see 3.2.1.a). The multiple grooves shall readily lead off leakage from any perforation of an inner tube to the end of the compartments formed between outer and inner tubesheets. Visual means shall be provided for readily detecting leakage through the use of a telltale system leading from the leakoff connections of each end compartment. An instruction plate in accordance with MIL-DTL-15024 and MIL-P-15024/5 shall be provided to indicate the function of these connections and to warn against improper plugging of them. This plate shall be located close to one of the leakoff connections. Inner tubes shall have one of the nominal outside diameters specified below, with corresponding wall thickness, and shall be sized for rolling into the outer tubesheets with a standard expander. After assembly, the specified tube-side hydrostatic test pressure shall be applied to the end compartments and leakage warning passages, and there shall be no leakage under this pressure.

3.3.9 Coatings. Paint, tinning, galvanizing, zinc coating, and plastic coating shall not be applied to any surface that will contact lubricating or hydraulic oil, such as the following examples:

a. Inside surface of shell subassembly.

b. Outside surface of tubes.

c. Support plates, baffles, spacers, tie rods, nuts, and locking devices.

d. Areas of tubesheets exposed to shell-side fluid.

3.3.9.1 Shell-side electroplating. For submarine battery cooling and for other applications as specified (see 6.2), the shell side shall be completely electroplated. Electroplating shall provide a thickness of 0.0010 to 0.0015 inch and shall have a smooth, fine grain appearance free from burnt deposits that may result from high current density in spots. Shell-side surfaces include the following:

a. Inside surface of shell subassembly.

b. Outside surface of tubes.

c. Support plates, baffles, spacers, tie rods, nuts, and locking devices.

d. Areas of tubesheets exposed to shell-side fluid.

3.3.10 Tubes. Tubes shall be of the following sizes and configurations.

3.3.10.1 Tube sizes. Unless otherwise specified (see 6.2), tube sizes shall be in accordance with 3.3.10.1.1 through 3.3.10.1.3.

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3.3.10.1.1 Submarine coolers, seawater coolant. Submarine coolers designed for use with seawater as the coolant shall be provided with tubes of ⅝-inch outside diameter, 0.065-inch (number 16 Birmingham wire gauge [BWG]) wall thickness.

3.3.10.1.2 Surface ship coolers, seawater coolant. Surface ship coolers designed for use with seawater as the coolant shall be provided with tubes of ⅝-inch outside diameter, 0.049-inch (number 18 BWG) wall thickness, or ½-inch outside diameter, 0.049-inch (number 18 BWG) wall thickness. The ⅝-inch outside diameter size shall be used for lubricating oil coolers for main propulsion turbine units that use scoop injection of circulating water.

3.3.10.1.3 Coolers with freshwater coolant. Coolers designed for use with fresh water as the coolant shall be provided with tubes of ⅝-inch outside diameter, 0.049-inch (number 18 BWG) wall thickness; ½-inch outside diameter, 0.049-inch (number 18 BWG) wall thickness; or ⅜-inch outside diameter, 0.035-inch (number 20 BWG) wall thickness.

3.3.10.2 Fins. Unless otherwise specified (see 6.2), fins may be used on the outer tube of double-tube coolers. The use of fins on single-tube coolers shall be as specified in 3.3.10.2.1 or 3.3.10.2.2.

3.3.10.2.1 Integral fins. Tubes shall be of the low-fin type with fins worked out of the tube wall in such a way that the outside diameter of the fins shall not exceed the outside diameter of the plain ends of the tubes. The tube wall at the plain end shall be two BWG sizes heavier than the wall thickness specified in 3.3.10.1 to permit use of a standard tube expander.

3.3.10.2.2 Applied fins. Tubes shall be of the low-fin type. One end of the finned tube shall be gradually enlarged (flared) to a diameter slightly greater than the outside diameter of the fins. This practice will permit removal and replacement of individual tubes. The enlarged end shall be the discharge end of the tube. The holes in the tubesheets for these enlarged tube ends shall be reamed 0.001 inch larger than the outside diameter of the tube end, with a +0.005-inch tolerance, and the nominal ligament between tube holes shall be not less than 3/16 inch.

3.3.10.3 Tube expansion. Tubes of coolers shall be expanded into the tubesheets, and the discontinuity in tube surface caused by the expansion shall be kept to a minimum. Unless otherwise specified (see 6.2), the expansion shall be by means of a tube expander in accordance with MIL-E-15809 and shall be governed by an automatic tube expander control. The expanded portion of the tube-to-tubesheet joint shall not approach closer than ⅛ inch to the shell side of the tubesheets, which includes the back face of single or outer tubesheets and either face of the inner tubesheets for double-tubesheet construction. Depths of expansion of tubes shall be not less than those specified in table XIV.

TABLE XIV. Minimum depth of tube expansion.

Tube outside diameter (inch)	Minimum depth of expansion (inch)
⅝	⅝
½	⅝
⅜	½

3.3.11 Tubesheets.

3.3.11.1 Minimum thickness. Tubesheet thickness shall be not less than the following:

a. For joints with inlet-end flare (see 3.3.11.3) – depth of expansion plus depth of flare plus ⅛ inch.

b. For nonflared joints – depth of expansion plus ⅛ inch.

c. For inner tubesheets – depth of expansion plus ¼ inch.

3.3.11.2 Spacing and size of holes. Holes for tubes in tubesheets shall be spaced center-to-center and drilled or reamed to the diameters shown in table XV.

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TABLE XV. Spacing and diameter of tube holes.

Tube outside diameter (inch)	Tube spacing center-to-center, to be not less than 1/ (inch)	Diameter of holes for tubes 2/ (other than where shell-side electroplating is specified) (inch)	Diameter of holes for tubes 2/ (where shell-side electroplating is specified) (inch)
5/8	13/16	0.626	0.631
1/2	21/32	0.503	0.508
3/8	1/2	0.376	0.381

NOTES:

1/ Nominal ligament thickness tolerance shall be +0.015 or -0.017 inch.

2/ The tolerance shall be -0.000 or +0.005 inch. A maximum of 1 percent (to the nearest hole) of the tube holes in each tubesheet shall range to +0.010 inch, provided that two oversized holes are not adjacent and that the ligament thickness requirements are met.

3.3.11.3 Tube end hole flaring. The holes for the ends of the tubes shall be flared as specified in table XVI to allow for belling the ends of the tubes.

TABLE XVI. Flaring of tube end holes.

Tube outside diameter (inch)	Radius of flare (inch)	Diameter of flared hole at the outside face of the tubesheet (inch)
5/8	1/2	3/4
1/2	3/8	5/8
3/8	5/16	1/2

3.3.11.4 Grooving of holes in tubesheets. Unless otherwise specified (see 6.2), the holes for tubes in the tubesheets shall be grooved with annular grooves as specified in table XVII. The groove design shall ensure that scratches to the tubes during installation either will not affect sealing capability or can be removed prior to tube expansion.

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TABLE XVII. Annular grooving of holes in tubesheets.

Tube outside diameter (inch)	Cooler application	Number of grooves in outer tubesheet	Number of grooves in inner tubesheet	Width of groove (minimum) (inch)	Depth of groove (±0.005 inch)	Space between grooves (inch)
⅝ and ½	Submarine cooler with single tubesheets and designed for submergence pressure	3	----	1/16	0.012	⅛
	Submarine cooler with double tubesheets and designed for seawater cooling	3	1	1/16	0.012	⅛
	Other coolers with single tubesheets	1	----	1/16	0.012	----
	Other coolers with double tubesheets	2	1	1/16	0.012	⅛
⅜	Double-tubesheet cooler	2	1	1/16	0.012	⅛
	Single-tubesheet cooler	1	----	----	----	----

NOTES:

1. With the double-tube construction specified in 3.3.8, no grooving of the inner sheet is required where a welded or brazed joint is used for the outer-tube-to-inner-tubesheet attachment.
2. The first groove of grooved outer tubesheets shall be ⅜ inch from the sheet face nearest the tube end.
3. The groove of grooved inner tubesheets shall be at the expanded portion of the joint.

3.3.11.5 Hole edge rounding. The edges of holes for tubes shall be rounded as specified in table XVIII, except where a flared contour is required for the tube inlet end.

TABLE XVIII. Rounding of tube hole edges.

Tube outside diameter (inch)	Edge of hole rounded to a radius of (inch)
⅝	1/16
½	1/16
⅜	1/16

3.3.11.6 Tubesheet flatness and finish. The sealing surface of the tubesheet(s) shall meet a flatness tolerance of 0.020 inch and the surface finish requirements (see 3.2.15.1) after assembly of the tube bundle.

3.3.12 Tube ends. The ends of tubes shall be expanded and belled. The inlet ends shall be finished flush with the face of the tubesheet. Discharge ends of tubes shall protrude not more than 1/16 inch beyond the face of the tubesheet but shall not terminate within a tubesheet.

3.3.13 Baffles. Within the shell space, close-fitting transverse baffles shall be installed to provide for multiple passes of the shell-side fluid across the tubes and to provide support for the tubes. The segmental type of baffle shall be used. Baffles shall have a finish of 250 microinches root mean square or smoother on the periphery, including the cut edge. The baffles shall be held in position by tie rods and spacer sleeves. The tie rods shall be threaded into, but not through, one of the tubesheets. Where one sheet is of the floating type, the tie rod shall be assembled to the stationary sheet. Nominal diameters of holes for tubes in baffles shall be not more than 1/64 inch greater than the outside diameter of the tube. The edges of holes shall have roughness removed.

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3.3.13.1 Baffle thickness. Minimum baffle thickness shall be in accordance with class C TEMA Standards except that where low-finned tubes are used, the baffle thickness shall be not less than ⅛ inch.

3.3.13.2 Baffle spacing. Baffles shall be spaced so that the unsupported tube spans (see 6.8.7) shall not exceed the values required in accordance with class C TEMA Standards.

3.3.13.3 Sealing strips. To improve cooler performance, sealing strips shall be installed between the tube bundle and the shell. These strips shall be installed in slots in the baffles and tack-welded.

3.3.14 Waterboxes. To allow distribution of coolant to the tubes, the waterbox depth measured normal to the tubesheet shall be not less than one-half the mean diameter of the tubesheet area exposed to the flow of the cooling water to the tubes. For coolers of equivalent inside shell diameter (D), the waterbox head depth shall be not less than 0.50 D for single-pass coolers, 0.345 D for two-pass coolers, or 0.25 D for four-pass coolers. For multipass coolers, preferred construction shall let coolant between passes flow either vertically upward or horizontally from pass to pass, with vent holes provided in the waterbox partitions.

3.3.14.1 Partition plate width. Waterbox pass partition plate width shall be not less than ¼ inch at the gasket contact surface.

3.3.14.2 Jackscrews. Jackscrews shall be fitted on waterboxes of coolers of 12-inch and larger nominal shell diameter.

3.3.14.3 Cathodic protection. Unless otherwise specified (see 6.2), readily replaceable zinc anodes in accordance with MIL-A-19521 shall be installed in the saltwater circuits of all coolers in accordance with the following:

a. For submarine submergence pressure applications, threaded support plugs shall not be used. Cover plates secured by four or more threaded fasteners shall be used for support of anodes.

b. For submarine non-submergence pressure applications, bolted cover plates or threaded plugs shall be used for support of anodes. Threaded plugs shall be in accordance with Drawing 803-5959186.

c. For surface ship applications, support plugs shall be of the straight-thread O-ring seal type in accordance with MIL-A-19521. In exception to the material requirements of MIL-A-19521, plug materials shall be in accordance with composition 70-30 copper-nickel alloy of MIL-C-15726.

3.3.14.4 Waterbox flatness and finish. The sealing surface(s) of the completed waterbox shall meet a flatness tolerance of 0.010 inch and the surface finish requirements specified in 3.2.15.1.

3.3.15 Cyclic life requirements for type I, class 1 coolers. Unless otherwise specified (see 6.2), cyclic life requirements for type I, class 1 coolers shall be as specified in 3.3.15.1 through 3.3.15.6.

3.3.15.1 Parts subject to submergence pressure. Parts, including structure composed of outer and inner tubesheets (together with short sections of tubes between paired inner and outer tubesheets), subject to submergence pressure shall be constructed in accordance with the ASME Boiler and Pressure Vessel Code, Section III, Nuclear Power Plant Component, class 1 components, as a minimum requirement, except that welding shall be in accordance with S9074-AR-GIB-010/278. Reinforcement of openings shall be integral with the waterbox shell or nozzle, or both; separate pads or saddle-type reinforcements shall not be used. Welds shall be ground smooth and the corners and fillets shall be rounded as necessary to minimize notch effect. In designing heat exchangers, the forces exerted by the circulating water inlet and discharge piping on their respective waterbox nozzles shall be calculated as follows:

$$\text{Piping axial load} = \pi\, r_m\, t\, (s_y - Pr_i/2t) \text{ pounds}$$

$$\text{Piping bending moment} = 1.3\pi\, r_m^2\, t\, (s_y - Pr_i/2t) \text{ pound-inches}$$

$$\text{Piping torsional moment} = \pi\, r_m^2\, t \sqrt{s_y^2 - (Pr_i/2t)^2} \text{ pound-inches}$$

Where (see 6.2):

P = design pressure for the component, lb/in^2

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r_i = pipe inside radius, inches

r_m = pipe mean radius, inches, = $(r_i + r_o)/2$

r_o = pipe outside radius, inches

s_y = tabulated value of yield strength of the piping material at the component's design temperature, lb/in^2

t = pipe thickness, inches

π = ratio of circumference to diameter, approximately 3.14159; and numerical coefficients and exponents are dimensionless

3.3.15.2 Cyclic pressure and external force loading. In the design of heat exchanger seawater sides, full consideration shall be given to the cyclic nature of the pressure loading and external force loading. Unless otherwise specified (see 6.2), the design shall be based on the following:

a. The number of total depth-changing cycles, with associated temperature variation, in conjunction with the piping reaction load (see 3.3.15.1) that produces the highest stress.

b. 150 cycles each of the specified hydrostatic test pressure applied in the waterbox or between the double tubesheets.

The analysis shall be in accordance with the methods and criteria of the ASME Boiler and Pressure Vessel Code, Section III, division 1, class 1 components. Design fatigue curves for materials commonly used in coolers and design stress intensity values for materials not covered in the ASME Boiler and Pressure Code, Section III, division 1, class 1 components, shall be obtained as specified (see 6.2). The method of analysis of flat perforated plates shall be in accordance with article A-8000 of the ASME Boiler and Pressure Vessel Code, Section III, division 1, appendices. The cumulative usage factor for all cyclic transients shall not exceed 0.8.

3.3.15.3 Bolting subject to submergence pressure. For type I, class 1 coolers, bolting subject to submergence pressure shall meet the following requirements:

a. Calculated bolt stress due to design pressure shall be limited to one-third of the yield strength of the material at the design temperature.

b. Maximum value of service stress averaged across the bolt cross-section and neglecting stress concentration factor shall be limited to two-thirds of the yield strength of the material at the design temperature.

c. Maximum value of service stress at the periphery of the bolt cross-section resulting from tension plus bending, but neglecting stress concentration factor, shall be limited to the yield strength of the material at the design temperature.

d. The range of coefficient of friction for the lubricant shall be accounted for in the determination of torque value as follows:

(1) Using the hydrostatic test load or the load based on design pressure plus maximum piping load on nozzles, whichever is greater, and high coefficient of friction ($\mu = 0.13$), determine installation torque using the following equation:

$$T = \frac{Pr_t}{12}\left[\frac{\cos\theta\ \tan\alpha + \mu}{\cos\theta - \mu\tan\alpha} + \frac{r_c}{r_t}\mu\right]$$

Where:

r_t = minimum pitch radius of external threads, inches

θ = ½ of thread angle, degrees

α = helix angle, degrees, = Arc tan $1/(2\pi r_t N)$

r_c = collar radius, average of $D_h/2$ and bolt major radius, inches

D_h = diameter across flat of nuts (average of maximum and minimum), inches

D = average of mean pitch and minor diameters of external thread, inches, = $2r_t - 0.32476/N$

μ = 0.13, coefficient of friction (high value), dimensionless

T = torque, foot-pounds

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N = number of threads per inch

P = hydrostatic test load or load based on design pressure plus maximum piping load on nozzles, whichever is greater, pounds

(2) Using the installation torque above and the low coefficient of friction ($\mu = 0.065$), determine S_{av}, the average stress on the bolt cross-section, from the following equation:

$$S_{av} = \frac{P}{6\pi\mu^2 {D_h}^2 D^3}\left[(D^2 + 16\mu^2 {D_h}^2)^{3/2} - D^3\right], \text{ lb/in}^2$$

P in this equation is the preload force, in pounds, found from the equation in (1) by substituting the low coefficient of friction ($\mu = 0.065$) and using the installation torque, T, calculated above. This calculated stress, when combined with other stresses specified in the ASME Boiler and Pressure Vessel Code, shall not exceed the limits in accordance with the ASME Boiler and Pressure Vessel Code.

e. Fatigue analysis of bolts shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section III, division 1, subsection NB, class 1 components, except that the cumulative usage factor shall not exceed 0.8.

f. Submergence pressure boundary studs and bolts shall be of a reduced-shank construction to reduce bending stresses.

3.3.15.3.1 Bolted waterbox inspection openings. Unless otherwise specified (see 6.2), threaded fasteners securing waterbox inspection openings subject to submergence pressure shall be constructed so that 200 inspection cover removal and reinstallation cycles, 150 hydrostatic test pressure cycles, and the number of total depth-changing cycles will not cause the maximum cumulative usage factor to exceed 0.8. The effect on threaded-fastener stress levels caused by movement between the inspection cover and inspection cover flange when the seawater side of the waterbox is pressurized shall be accounted for when calculating the maximum cumulative usage factor. The amount of movement or relative slip used in the foregoing analysis shall be confirmed during factory hydrostatic testing (see 4.3.3). The fastener design shall be in accordance with the ASME Boiler and Pressure Vessel Code, Section III, division 1, subsection NB, class 1 components, when the measured slip (rather than the assumed slip) is factored into the analysis.

3.3.15.4 Number of seawater passes. Coolers subject to submergence pressure shall have an even number of seawater passes so that the inlet and outlet seawater nozzles will be on the same waterbox. The stiffness of the piping shall not affect expansion or contraction of the cooler tubes.

3.3.15.5 Double-tubesheet design.

3.3.15.5.1 Assembly of double tubesheets. For type I, class 1 coolers, the double-tubesheet design shall be either bolted and gasketed with an O-ring or completely welded.

3.3.15.5.1.1 Bolted and gasketed, with O-ring. The outer tubesheet shall be secured to the inner tubesheet or spacer piece by means of threaded fasteners. These fasteners shall be of sufficient size and number to maintain tightness of the O-ring gasketed joint whenever specified design pressure is applied to the space between double tubesheets without dependence on the waterbox flange bolting. The O-ring gasket shall be replaceable without the necessity of retubing the cooler. The spacer piece, if provided, shall be welded (full penetration weld) to the inner tubesheet. The weld shall be in accordance with class A-F of S9074-AR-GIB-010/278, except that for magnetic material, the weld may be magnetic-particle inspected on the root passes, on the back chipped side of the initial root weld, and on the surface passes in accordance with S9074-AR-GIB-010/278 in lieu of radiographic inspection. If materials involved are not magnetic, then liquid-penetrant inspection of the root passes, the back chipped side of the initial root weld, and the surface passes in accordance with S9074-AR-GIB-010/278 may be substituted for radiographic inspection. When specified (see 6.2), the structural adequacy of this welded joint shall be demonstrated by analysis.

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3.3.15.5.1.2 Completely welded. The outer tubesheet shall be secured to the inner tubesheet by a full penetration butt-welded joint. This joint shall maintain the tightness of the assembly when the specified design pressure is applied to the space between the sheets without dependence on the waterbox flange bolting. This weld shall be in accordance with class A-F of S9074-AR-GIB-010/278, except that for magnetic material the weld may be magnetic-particle inspected on each weld pass in accordance with S9074-AR-GIB-010/278 in lieu of being radiographed. If materials involved are not magnetic, liquid-penetrant inspection of each weld pass in accordance with S9074-AR-GIB-010/278 shall be substituted for magnetic-particle inspection.

3.3.15.5.2 Tubes for type I, class 1 coolers. For type I, class 1 coolers, tubes shall be rolled using an approved procedure as specified (see 6.2 and A.3.1.1) and a mockup of the inter-tubesheet joint shall be tested for tensile strength as specified (see 6.2 and appendix A). When specified (see 6.2 and A.3.1.5), a tube-to-tubesheet joint quality assurance plan shall be implemented to provide high confidence that the joint will meet the cyclic life requirements specified.

3.3.15.6 Waterbox. Waterboxes for cyclic loading shall be hemispherically shaped with nozzles aligned radially. Waterboxes not constructed to permit analysis for fatigue life adequacy shall operate as specified when tested in accordance with 4.2.

3.4 Type II (fabricated tube designed) requirements.

3.4.1 Materials.

3.4.1.1 Type II, class 2. Type II, class 2 coolers (submarine, seawater cooled, seawater side secure at depths greater than 200-foot submergence) shall be constructed of the materials specified in table XIX.

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TABLE XIX. Type II, class 2 cooler materials.

Part	Material	Applicable document
Casings, fabricated	Copper-nickel alloy, composition 90-10	MIL-C-15726
	Tubing, copper-nickel alloy, composition 90-10	MIL-T-16420
Casings, cast, and core frames	Copper-nickel alloy, composition C96200	ASTM B369/B369M
	Tin bronze, alloy C90300	
Header plates, baffles, and reinforcement plates	Copper-nickel alloy, composition 90-10	MIL-C-15726
Tube halves	Copper-nickel alloy 1/	MIL-C-15726
Oil tube centers	Steel, cold rolled	Commercial
Brazing sheets	Electrolytic copper	Commercial
Covers, tube-side	Copper-nickel alloy, composition 90-10	MIL-C-15726
	Bronze, tin, sand casting, alloy C92200 or C90300	ASTM B584
Pipe plugs and adapters	Copper-nickel alloy, composition 70-30	MIL-C-15726
	Nickel-copper alloy	QQ-N-281
Packing rings 2/	Rubber, synthetic, class 2 or 5	MIL-PRF-1149
Flat gaskets	Non-asbestos-containing material	MIL-DTL-24696
Bolts or studs, seawater-side	Nickel alloy, grade 400 or 500	MIL-DTL-1222
Nuts for bolts and studs, seawater-side	Nickel alloy, grade 400 or 500	MIL-DTL-1222
Bolts, studs, and nuts for service other than as specified above	Copper alloy, grade 651, 632, or nickel alloy, grade 400 or 405	MIL-DTL-1222
	Copper-aluminum alloy, copper alloy C61400	QQ-C-465
Lockwashers	Phosphor bronze or silicon bronze	ASME B18.21.1
Jackscrews	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
	Phosphor bronze	ASTM B139/B139M
	Copper-silicon alloy	ASTM B98/B98M

NOTES:

1/ Tube halves shall be formed from a bimetal strip made of an outer layer of composition 90-10, 0.006 to 0.009 inch thick, clad on a basic layer of composition 70-30. Thickness of the strip shall be not less than 0.020 inch.

2/ See 3.2.25.

3.4.1.2 Type II, class 3. Type II, class 3 coolers (surface ship, seawater cooled) shall be constructed of the materials specified in table XX.

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TABLE XX. Type II, class 3 cooler materials.

Part	Material	Applicable document
Casings, fabricated	Copper-nickel alloy, composition 90-10	MIL-C-15726
	Tubing, copper-nickel alloy, composition 90-10	MIL-T-16420
Casings, cast, and core frames	Copper-nickel alloy, composition C96200	ASTM B369/B369M
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
Header plates, baffles, and reinforcement plates	Copper-nickel alloy, composition 90-10	MIL-C-15726
Tube halves	Copper-nickel alloy 1/	MIL-C-15726
Oil tube centers 2/	Steel, cold rolled	Commercial
Brazing sheets	Electrolytic copper	Commercial
Covers, tube-side	Copper-nickel alloy, composition 90-10	MIL-C-15726
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
	Valve bronze	ASTM B61
	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
		ASTM B169/B169M
Packing rings 3/	Rubber, synthetic, class 2 or 5	MIL-PRF-1149
Flat gaskets	Non-asbestos-containing material	MIL-DTL-24696
Bolts, studs, nuts, and jackscrews	Copper alloys	MIL-DTL-1222
Lockwashers	Phosphor bronze or silicon bronze	ASME B18.21.1

NOTES:

1/ Tube halves shall be formed from a bimetal strip made of an outer layer of composition 90-10, 0.006 to 0.009 inch thick, clad on a basic layer of composition 70-30. Thickness of the strip shall be not less than 0.020 inch.

2/ Where nonmagnetic construction is specified, oil tube centers shall be 70-30 copper-nickel alloy in accordance with MIL-C-15726.

3/ See 3.2.25.

3.4.1.3 Type II, class 4. Type II, class 4 coolers (submarine, freshwater cooled) shall be constructed of the materials specified in table XXI.

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TABLE XXI. Type II, class 4 cooler materials.

Part	Material	Applicable document
Tube halves	Copper-nickel alloy no. 710	ASTM B122/B122M
All parts not otherwise specified	Copper-nickel alloy	MIL-C-15726
	Copper-nickel alloy, wrought: bars, forgings, plates, rods, shapes, sheets, and strips	QQ-C-591
	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
	Copper	ASTM B152/B152M
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
	Valve bronze	ASTM B61
	Copper-nickel alloy, cast, composition C96200	ASTM B369/B369M
Oil tube centers	Steel, cold rolled	Commercial
Brazing sheets	Electrolytic copper	Commercial
Packing rings [1/]	Rubber, synthetic, class 2 or 5	MIL-PRF-1149
Flat gaskets	Non-asbestos-containing material	MIL-DTL-24696
Bolts, studs, nuts, and jackscrews	Nickel alloys or copper alloys	MIL-DTL-1222
Lockwashers	Phosphor bronze or silicon bronze	ASME B18.21.1
NOTE: [1/] See 3.2.25.		

3.4.1.4 Type II, class 5. Type II, class 5 coolers (surface ship, freshwater cooled) shall be constructed of the materials specified in table XXII.

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TABLE XXII. Type II, class 5 cooler materials.

Part	Material	Applicable document
All parts not otherwise specified	Copper-nickel alloy	MIL-C-15726
	Copper-silicon alloy, wrought: bars, forgings, plates, rods, shapes, sheets, and strips	QQ-C-591
	Copper-aluminum alloy, copper alloy C61400	ASTM B150/B150M
	Copper	ASTM B152/B152M
	Bronze, tin, sand castings, alloy C90300 or C92200	ASTM B584
	Valve bronze	ASTM B61
	Copper-nickel alloy, cast, composition C96200	ASTM B369/B369M
Oil tube centers 1/	Steel, cold rolled	Commercial
Brazing sheets	Electrolytic copper	Commercial
Packing rings 2/	Rubber, synthetic, class 2 or 5	MIL-PRF-1149
Flat gaskets	Non-asbestos-containing material	MIL-DTL-24696
Bolts, studs, nuts, and jackscrews	Nickel alloys or copper alloys	MIL-DTL-1222
Lockwashers	Phosphor bronze or silicon bronze	ASME B18.21.1

NOTES:

1/ Where nonmagnetic construction is specified, oil tube centers shall be copper-nickel alloy in accordance with MIL-C-15726.

2/ See 3.2.25.

3.4.1.5 Use of aluminum alloys. When specified (see 6.2), type II coolers shall be constructed entirely of aluminum alloys.

3.4.1.6 Nonmagnetic material. Where nonmagnetic construction is specified (see 6.2), all material, except the special tube halves, shall be 70-30 copper-nickel.

3.4.2 Minimum thickness. Thickness of header plates shall be not less than 0.090 inch. Thickness of tube halves shall be not less than 0.020 inch.

3.4.3 Assembling of strut- and plate-type tubes. Strut- and plate-type tubes shall be assembled by furnace brazing formed tube halves that have been mechanically assembled with brazing sheets. Grid-type centers shall be provided for oil tubes to improve flow characteristics.

3.4.4 Forming larger assemblies.

3.4.4.1 Larger assemblies from strut-type tubes. Unless otherwise specified (see 6.2), the assembling of strut-type tubes into cores, case and core subassemblies, or coolers shall be accomplished by either furnace brazing or lead-free silver soldering.

3.4.4.2 Larger assemblies from plate-type tube. The assembling of plate-type tubes into cores shall be accomplished by furnace brazing.

3.4.5 Casings and covers. Casings and covers shall be of brazed, welded, or cast construction, except that brazed construction shall not be used for the casings of class 2 coolers.

3.5 Workmanship. Workmanship shall be in accordance with the requirements of approved manufacturer drawings.

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3.6 Cleanliness practices. In addition to any special cleaning and cleanliness requirements specified (see 6.2), the following practices shall be followed, as a minimum:

a. The external surfaces of each tube shall be cleaned prior to assembly of the tube bundle. Shell or casing internal surfaces and tube bundle external surfaces shall be clean and dry when the tube bundle is installed in the shell or casing. The above cooler surfaces shall be free of corrosion, burrs, chips, dust, oil, grit, scale, and other foreign material when assembled.

b. Subassemblies, such as tube bundles, shells or casings, and waterboxes, shall be covered while not undergoing an active procedure during the fabrication process.

c. Completed subassemblies awaiting assembly shall be stored in a clean area or otherwise protected from contaminants.

d. All openings into completed units, such as nozzles, shall be covered with pads or otherwise sealed to prevent the introduction of contaminants and water.

e. When specified (see 6.2), fluid used for testing and flushing shall meet the specified cleanliness requirements.

3.7 Vibration. Coolers shall be designed to meet the vibration requirements of 3.7.1 and 3.7.2.

3.7.1 Environmental vibration. Coolers shall be designed to withstand external shipboard environmental vibration in accordance with MIL-STD-167-1, type I without damage to or loosening of parts or degradation of operational performance under the specified design conditions (see 3.2.1).

3.7.2 Flow-induced vibration. Coolers shall be designed to withstand the peak shell-side flow rate (see 3.2.1) without damage, fatigue failure, or loosening of parts under flow-induced vibrations.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the performance test, pressure drop test, high-impact shock test, and vibration analysis specified in 4.2.1 through 4.2.3 and 4.2.6, the tests specified in 4.3.3, and the examinations specified in 4.3.1 and 4.3.2. For applications for which cyclic life requirements are established, design adequacy of type I, class 1 coolers shall be verified by either analysis (see 4.2.4) or test (see 4.2.5). Test conditions for the performance and pressure drop tests shall be as specified (see 6.2).

4.2.1 Performance test. The cooler shall be subjected to a factory or laboratory test to demonstrate the ability of the cooler to remove the required heat capacity under design conditions of flow, temperature, and pressure of the cooled and cooling mediums as specified (see 3.2.1).

4.2.2 Pressure drop test. The cooler shall be tested under the flow conditions as specified (see 6.2) in both cooled and cooling mediums to ensure the allowable pressure drop requirements of 3.2.10 are met.

4.2.3 High-impact shock test. Unless otherwise specified (see 6.2), the high-impact shock test shall be as specified for grade A, class I in accordance with MIL-DTL-901 and as follows:

a. One cooler of each size and similar construction shall be shock tested.

b. The cooler shall be shock tested in its installed position.

c. The shell and tube sides of the cooler shall be pressurized to design operating pressures during the test. Equipment or parts shall be mounted on the shock machine in a manner simulating the installation aboard ship with the shipboard mounting fixtures being used. (For barge-tested units, barge foundation shall be shipbuilder furnished.) The weight designation of the shock test shall be as required by the combined weight of the equipment being tested. When the cooler is being tested, grade 2 bolts in accordance with MIL-DTL-1222 shall be used to hold down the cooler during the shock test.

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4.2.3.1 Failure to perform principal function. Any of the following failures shall be considered cause for rejection of the equipment:

a. Breaking of any part.

b. Appreciable distortion or dislocation of any parts such as mounting feet, tubes, baffles, or attached piping.

c. Leakage of water from the cooling water side of the cooler. Following the shock test, the cooler shall be hydrostatically tested in accordance with 4.3.3 and shall show no evidence of leakage.

4.2.3.2 Disposal of shock-tested equipment.

4.2.3.2.1 Non-passing equipment. Equipment that has been subjected to the high-impact shock test but has failed to perform any of the principal functions specified herein shall not be accepted, either in whole or in any of its parts.

4.2.3.2.2 Passing equipment. Equipment that has been subjected to the high-impact shock test and has successfully performed all the principal functions specified herein shall not be considered for acceptance until it is first disassembled, inspected, and refurbished.

4.2.4 Design adequacy of type I, class 1 coolers. For applications for which cyclic life requirements are established, an analysis demonstrating the adequacy of the design to the requirements of 3.3.15.1 through 3.3.15.6 shall be accomplished. Appendix A provides guidance to procuring authorities for report contents (see 6.2).

4.2.5 Proof test – special case (cyclic loading stipulated) for type I, class 1 coolers. When specified (see 6.2), for the special case of an inlet-outlet waterbox not amenable to analysis for determination of its adequacy for meeting cyclic life requirements, the adequacy of the waterbox or a scale model thereof shall be verified by a proof test. This proof test shall consist of a qualitative survey of the waterbox utilizing either brittle coating (stress coat or its equivalent) or photostress (molded birefringent coatings) techniques in order to determine the location of maximum strain. Quantitative measurement of the magnitude of maximum principal strain in the waterbox shall be obtained from strain gauges of ⅛-inch gauge length or shorter. Each waterbox loading shall comprise a combination of the design submergence pressure with a piping reaction load (the load that will produce the highest stress) applied to the nozzles. The nozzle loads shall be calculated as specified in 3.3.15.1. A loading jig utilizing mechanical or hydraulic jacks shall be provided to apply external loads to the nozzle flanges. The following two combinations of loads shall be applied:

a. Internal pressure and external piping load (the piping reaction load that produces the highest stress) applied to one flange.

b. Internal pressure and mutually equal external loads (the piping reaction load that provides the highest stress) applied to both flanges.

4.2.5.1 Evaluation. When the maximum value of principal strain (stress) in the waterbox has been determined for the load combinations specified in 4.2.5.a or 4.2.5.b, the following steps shall be taken in evaluating design:

a. Determine the number of allowable cycles (corresponding to the maximum applied stress) from the design fatigue curve for the particular material involved. Call this number of cycles N_1.

b. Unless otherwise specified (see 6.2), determine the following:

(1) The number of design cycles, n_1, corresponding to 100 percent test depth changes

(2) The number of design cycles, n_2, corresponding to 50 percent test depth changes

(3) The number of design cycles, n_3, corresponding to 150 hydrostatic test pressure cycles.

c. Determine the number of allowable cycles (corresponding to half the maximum applied stress) from the design fatigue curve for the particular material involved. Call this number of cycles N_2.

d. Determine the number of allowable cycles (corresponding to 1.5 times the maximum applied stress) from the design fatigue curve for the particular material involved. Call this number of cycles N_3.

e. Form the sum of ratios, $n_1/N_1 + n_2/N_2 + n_3/N_3$. If this sum does not exceed 0.8, the waterbox design meets the criterion for resistance to fatigue.

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4.2.6 Vibration analysis. Cooler designs shall be analyzed to ensure compliance with the environmental vibration and flow-induced vibration requirements of 3.7.1 and 3.7.2, respectively. Vibration testing is not required.

4.2.6.1 Environmental vibration analysis. Coolers shall be verified to meet the type I environmental vibration criteria of MIL-STD-167-1 to ensure compliance with the requirements of 3.7.1. Environmental vibration adequacy shall be demonstrated by analysis with the natural frequencies of all component parts and appurtenances verified to not exceed the specified frequency (see 6.2).

4.2.6.2 Flow-induced vibration. Coolers shall be verified to meet the requirements of 3.7.2. Flow-induced vibration adequacy shall be demonstrated by analysis following the specified method (see 6.2). For coolers where the fluid to be cooled is on the shell-side of the cooler, the specified peak flow rate (see 3.2.1) shall be considered a steady-state flow rate for this analysis. For coolers where the fluid to be cooled is on the tube-side, the maximum shell-side flow rate (see 3.2.1) shall be used. This analysis shall address susceptibility to flow-induced vibrations of all component parts and appurtenances on the shell-side of the cooler, such as the tubes, tie rods, tie rod spacers, and baffles.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.3.1 and 4.3.2 and the tests of 4.3.3.

4.3.1 Examination. Each cooler offered for delivery shall be examined for adjustment, fit, material, finish, workmanship, and any other requirements specified herein not involving tests. The list of materials used shall be inspected to verify that prohibited materials have not been used (see 3.2.5). Cooler components and subassemblies shall be visually inspected for cleanliness to the requirements of 3.6.

4.3.2 Critical dimensions. For type I, class 1 coolers, critical dimensions (see 6.8.2) such as waterbox thicknesses, tubesheet thicknesses, flange thickness, tube hole sizes, and tubesheet ligaments shall be measured. Departures from limits established by accepted drawings shall be classified as major non-conformances. For type I, class 1 coolers, a 100 percent examination of the tubesheet holes and ligament widths shall be made.

4.3.3 Hydrostatic tests. Each cooler offered for delivery shall be given a separate hydrostatic test of the shell side, tube side, and void spaces (between double tubesheets), as applicable. Unless otherwise specified (see 6.2), test pressures shall be held for not less than 30 minutes, and leakage shall not occur. Additional requirements shall be as specified in 4.3.3.1 and 4.3.3.2.

4.3.3.1 Cooler test pressure. Unless otherwise specified (see 6.2), coolers shall be tested at 1½ times design pressure. Double-tubesheet void spaces and leakage warning passages shall be tested to the same pressure as the coolant side.

4.3.3.2 Additional verifications for type I, class 1 coolers. For type I, class 1 coolers, a hydrostatic test of the assembled cooler shall be performed. When the specified hydrostatic test pressure is reached, the pump and any other pressurizing devices shall be isolated from the cooler under test. The test pressure shall be applied for not less than 2 hours, with all of the joints and connections inspected after 2 hours with test pressure still applied. Any leakage shall be cause for rejection. The tubesheets shall be inspected during the hydrostatic test of the shell side to ensure that no leakage or weepage from the tube or tube joint occurs. During hydrostatic testing of the double-tubesheet void space, any drop in pressure greater than that expected as a result of temperature changes shall be cause for rejection of the cooler.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be in accordance with the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coolers described in this specification are intended for use in lubrication, hydraulic, and freshwater systems aboard Navy surface ships and submarines for the cooling of hydrocarbon-based oils and fresh water. No one cooler is intended to cool more than one of these three categories of fluid in any one application. The coolers are divided into two types according to their design and to whether the fluid to be cooled is the inner fluid or outer (see 1.2). The coolers are also divided into five classes according to whether they are for use on submarines or surface ships, whether they use seawater or fresh water as a cooling medium, and whether or not they are to be subjected to seawater pressure at depths greater than 200 feet (see 1.2). Coolers of type I may be of any class; coolers of type II may be of any class except class 1. Current stock of the coolers may be used until depleted; future requisitions for the coolers should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Type and class of cooler (see 1.2).

c. Intended cooler application (see 1.3).

d. When first article inspection is required (see 3.1).

e. Identification of fluid to be cooled (lubricating oil should be designated by specification or Military symbol, when applicable) (see 3.2.1.a).

f. Design and peak flow rates of fluid to be cooled (gal/min) (see 3.2.1.a).

g. Inlet and outlet temperatures of fluid to be cooled (°F) (see 3.2.1.a).

h. Design pressure of fluid to be cooled (lb/in^2) (see 3.2.1.a).

i. Required rate of heat transfer (British thermal units [Btu] per hour) (see 3.2.1.a).

j. Identification of coolant (see 3.2.1.b).

k. Design flow rate of coolant (gal/min) (see 3.2.1.b).

l. Inlet and outlet temperatures of the coolant (°F) (see 3.2.1.b).

m. Design pressure of coolant (lb/in^2) (see 3.2.1.b).

n. Space and weight limitations governing the cooler size, if established (see 3.2.2).

o. Ship motion and attitude requirements, if other than specified (see 3.2.3).

p. Allowable design point pressure drops for respective cooler sides in lb/in^2, if other than as specified (see 3.2.10).

q. Inlet temperature of coolant, if other than specified (see 3.2.12).

r. Inlet and outlet connection size and whether hose connections are required (see 3.2.14).

s. Requirements of inlet and outlet flanged connections, if other than specified (see 3.2.15).

t. Equivalent specification or Project Peculiar Document governing connection flange requirements for submarines (see 3.2.15.a).

u. Size, wall thickness, and material of installer-furnished vent and drain tubing (see 3.2.16).

v. Whether pressure gauge connections are required at inlet and outlet of coolant and cooled-fluid sides (see 3.2.17).

w. Foundation or parent machine interface requirements for cooler supports (see 3.2.18).

x. When O-ring gaskets are required in lieu of flat gaskets (see 3.2.22).

y. Gasket thickness, if other than specified (see 3.2.23).

z. Requirements for type I cooler gasket material when metal-to-gasket temperature is greater than 250 °F, if other than specified (see 3.2.23.b).

aa. Requirements for O-ring gaskets, if other than specified (see 3.2.24).

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bb. When a blank entry of four spaces for the unit number should be provided on the identification plate (see 3.2.26.m).

cc. Critical material identification and control requirements (see table V, note 6).

dd. When additional material requirements or restrictions apply to type I, class 4 and 5 coolers (see 3.3.2.4 and 3.3.2.5).

ee. Materials for the shell side of coolers in lubricating and hydraulic oil cooling applications, if other than specified (see 3.3.2.6.1).

ff. Whether aluminum alloys in table XI may be used for the shell-side internals of type I oil coolers (see 3.3.2.6.2).

gg. Whether the titanium and steel materials in table XII may be used for type I, class 3 coolers servicing lubricating or hydraulic oil (see 3.3.2.6.3).

hh. When U-bend tubes are required (see 3.3.3).

ii. Tube bending procedure reporting and approval requirements (see 3.3.3.1, 3.3.7, and A.3.1.4).

jj. When special construction consisting of a coolant flow path led through the shell with coolant piping connections integral with the shell should be used (see 3.3.5.2.c).

kk. When double-tubesheet construction is required for other than hydraulic oil coolers (see 3.3.7).

ll. When double-tube construction is required (see 3.3.8).

mm. When applications other than submarine battery cooling require shell-side electroplating (see 3.3.9.1).

nn. Outside diameter and wall thickness of tubes, if other than as specified (see 3.3.10.1).

oo. When fins are not to be used on the outer tube of double-tube coolers (see 3.3.10.2).

pp. Method of tube expansion, if other than specified (see 3.3.10.3).

qq. Annular grooving design of holes in tubesheets, if other than specified (see 3.3.11.4).

rr. When cathodic protection is not required in the saltwater circuit of the cooler (see 3.3.14.3).

ss. Cyclic life requirements for type I, class 1 coolers, if other than specified (see 3.3.15 and 4.2.5.1.b).

tt. Diameter and thickness of attached circulating water piping (see 3.3.15.1).

uu. The number of design cycles corresponding to 100 percent test depth changes and the number of design cycles corresponding to 50 percent test depth changes for type I, class 1 coolers (see 3.3.15.2, 3.3.15.3.1, and 4.2.5.1).

vv. Design fatigue curves for materials commonly used in coolers and design stress intensity values for materials not covered in the ASME Boiler and Pressure Code, Section III, division 1, class 1 components (see 3.3.15.2).

ww. Threaded fasteners construction maximum cumulative usage factor, if other than specified (see 3.3.15.3.1).

xx. When and how the structural adequacy of the spacer piece between tubesheets is to be demonstrated (see 3.3.15.5.1.1).

yy. Tube rolling procedure reporting and approval requirements for type I, class 1 coolers (see 3.3.15.5.2 and A.3.1.1).

zz. Tensile test procedure requirements for type I, class 1 coolers (see 3.3.15.5.2 and A.3.1.2).

aaa. When a tube-to-tubesheet joint quality assurance plan is required for type I, class 1 coolers (see 3.3.15.5.2 and A.3.1.5).

bbb. When aluminum alloys are specified for type II coolers (see 3.4.1.5).

ccc. When nonmagnetic construction is specified for type II coolers (see 3.4.1.6).

ddd. Requirement to seal the tube-to-header joints of type II coolers with strut-type tubes, if other than specified (see 3.4.4.1).

eee. Additional cleaning and cleanliness requirements for type I coolers, if specified (see 3.6).

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fff. When there are cleanliness requirements for fluid used for testing and flushing (see 3.6.e).

ggg. Test conditions for performance and pressure drop testing (see 4.2, 4.2.1, and 4.2.2).

hhh. High-impact shock test requirements, if other than specified (see 4.2.3).

iii. Analysis of design adequacy reporting requirements for type I, class 1 coolers (see 4.2.4 and A.3.1.3).

jjj. When a proof test is required for type I, class 1 coolers with the special case of inlet-outlet waterbox not amenable to analysis (see 4.2.5).

kkk. Maximum frequency for environmental vibration analysis (see 4.2.6.1).

lll. Method for performing flow-induced vibration analysis (see 4.2.6.2).

mmm. Duration of time for which pressure is to be held during the hydrostatic test, if other than specified (see 4.3.3).

nnn. Hydrostatic test pressures for cooler shell sides, tube sides, and void spaces, as applicable, if other than specified (see 4.3.3.1).

ooo. Packaging requirements (see 5.1).

ppp. Requirements for drawings (see 6.3.1 and appendix B).

qqq. Requirements for technical manuals (see 6.3.2 and appendix C).

rrr. Recommended data item descriptions (DIDs) for data deliverables (see 6.3.3).

sss. Requirements for provisioning technical documentation (PTD), spare parts, and repair parts (see 6.5).

ttt. Preservation level A or commercial (see D.3.1).

uuu. Oil- and water-side preservatives (see D.3.1.1.2.2).

vvv. Packing level A, B, C, or commercial (see D.3.2).

www. Whether lumber and plywood used for packaging should be fire-retardant treated material (see D.3.2.1).

xxx. Special marking, when required (see D.3.3).

6.3 Consideration of data requirements. The following guidance should be considered when this specification is applied on a contract.

6.3.1 Drawings. When specified (see 6.2), appendix B provides guidance to procuring authorities for the development of drawings.

6.3.2 Technical manuals. When specified (see 6.2), appendix C provides guidance to procuring authorities for the development of technical manuals.

6.3.3 Data deliverables. As specified (see 6.2), the Technical Authority (see 6.8.6) should be contacted for recommended DIDs for data deliverables.

6.4 Packaging guidance. Guidance for preservation and packaging is contained in appendix D.

6.5 Provisioning. PTD, spare parts, and repair parts should be furnished as specified (see 6.2). Spare parts and repair parts should meet the same requirements as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Cross-reference. Type I and II coolers were formerly known as type A and B, respectively. This change in designation does not affect storability or usability. Old stocks need not be altered merely to change type designation.

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6.8 Definitions.

6.8.1 Bottoming. Bottoming is screwing a screw or stud into a tapped hole until it contacts the bottom of the hole before all required threads are engaged with the hole. On screws that are torqued, bottoming can result in failing to achieve the target preload on a fastener that is being torqued even though the target torque is reached. Bottoming should be avoided since it can generate large forces at the contact point, which can crack the material.

6.8.2 Critical dimensions. A critical dimension is a dimension based on fatigue analysis or directly related to another dimension so based (for example, the size of a tube hole).

6.8.3 Design pressure. The pressure used in calculating minimum wall thicknesses of piping and piping components.

6.8.4 Front end (type I cooler). The front end is the end at which the coolant first arrives in normal operation. The opposite end is the rear end. The terms “front” and “rear” relate to the cooler, while “forward”, “aft”, and “after” relate to the ship in which the cooler is installed.

6.8.5 Shouldering. Shouldering occurs when the thread runout at the beginning of the unthreaded shank enters the threaded hole, causing local distortion of the threads and surrounding area.

6.8.6 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.8.7 Unsupported tube length. The unsupported tube length is the distance measured along the outer edge of the tube from one support to the next support by means of the U-bend for baffles adjacent to the bends of a U-bend-tube exchanger.

6.9 Subject term (key word) listing.

Drums

Fins

Flanges

Gaskets

Heat exchanger

Resistance, foundry

Shells

Tubes

Tubesheets

Vents

6.10 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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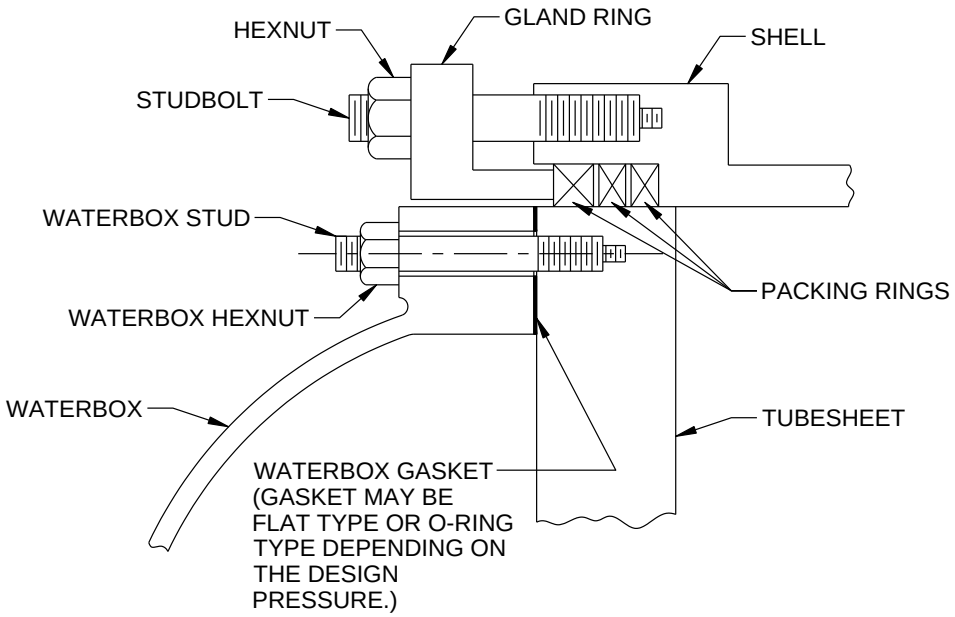


FIGURE 1. Straight tube, floating head, removable bundle, outside packed stuffing box.

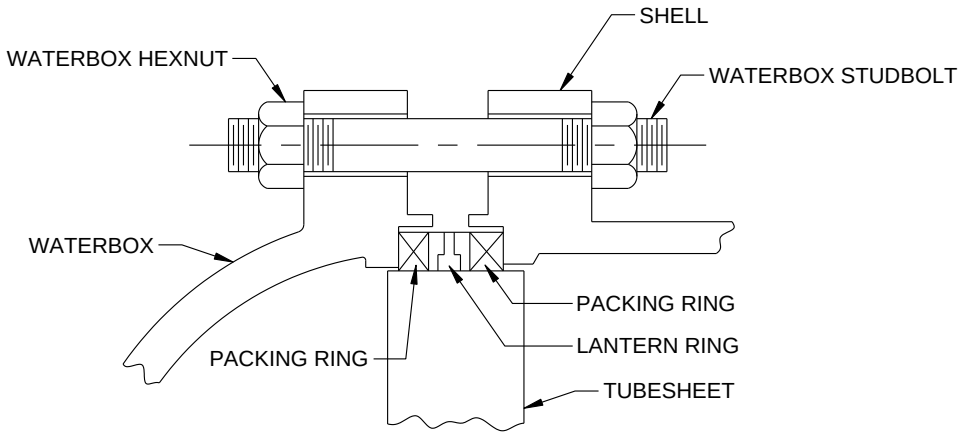


FIGURE 2. Straight tube, floating head, removable bundle, outside packed lantern ring construction.

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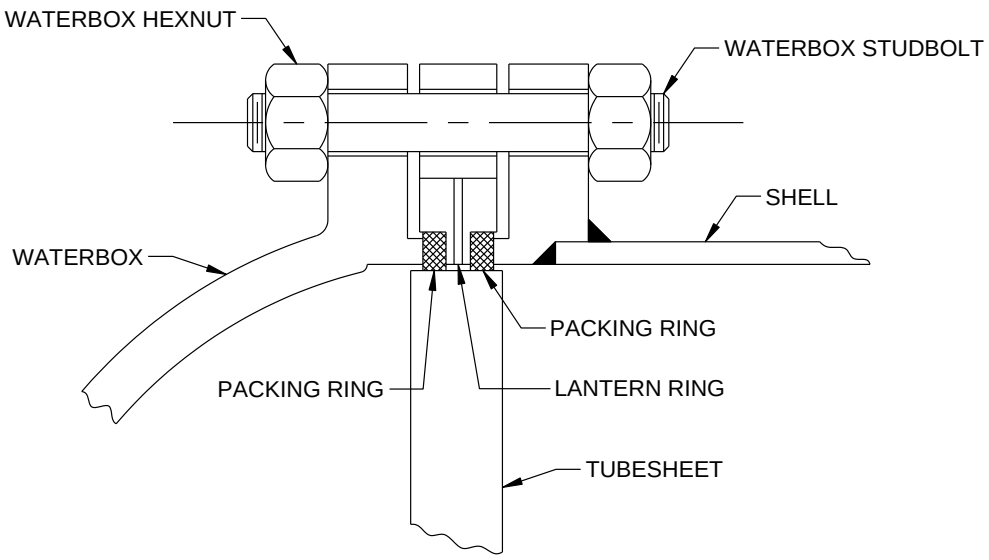


FIGURE 3. Straight tube, floating head, removable bundle, outside packed lantern ring construction, waterbox studbolt threaded into lantern ring.

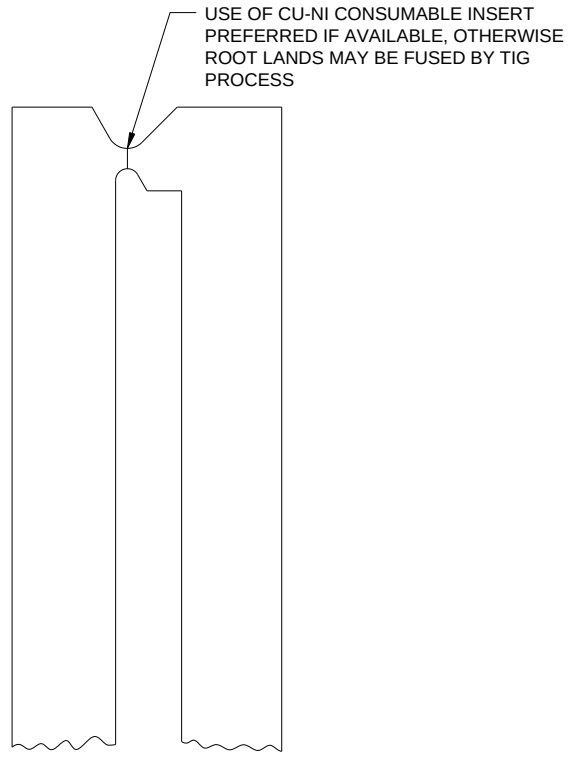


FIGURE 4. Acceptable method of welding inner and outer tubesheets – all double tubesheet cooler applications.

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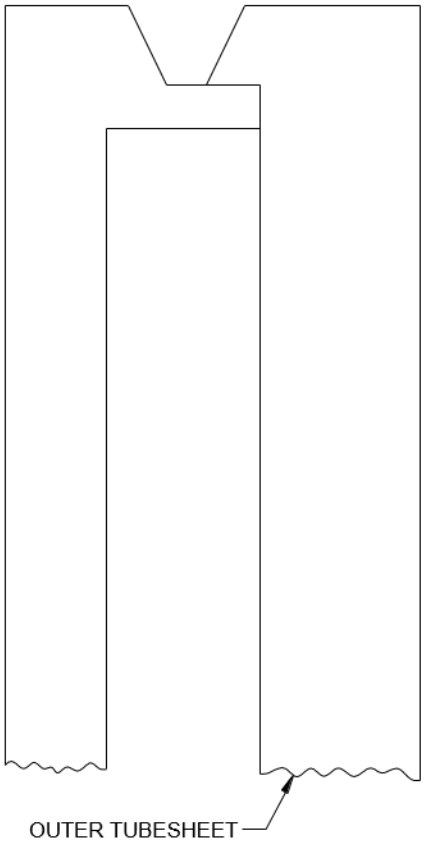


FIGURE 5. Acceptable method of welding inner and outer tubesheets – cooler applications for which cyclic life requirements are not specified.

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GUIDANCE FOR TEST REPORTING TECHNICAL CONTENT

A.1 SCOPE

A.1.1 Scope. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only. This appendix provides procuring activities guidance on information that should be specified by the acquisition documents for inclusion in the reports.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1627 - Bending of Pipe or Tube for Ship Piping Systems

(Copies of this document are available online at https://quicksearch.dla.mil/.)

A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B395/B395M - Standard Specification for U-Bend Seamless Copper and Copper Alloy Heat Exchanger and Condenser Tubes

(Copies of this document are available online at www.astm.org.)

A.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

A.3 SUGGESTED TEST REPORT CONTENTS

A.3.1 Report contents. The reports should contain, but are not limited to, the following information.

A.3.1.1 Tube rolling procedure. For type I, class 1 coolers, a tube rolling procedure should be developed and submitted to the command or agency concerned for approval. This procedure should be supported by data derived from a mockup test to demonstrate the adequacy of the tube rolling procedure. Rolled tubes and tubesheets in cooler assemblies used for first article inspection are to be manufactured, processed, and assembled (i.e., machined, lubricated, cleaned, and inspected) using the same techniques used in production. The procedure should include the following:

a. A sketch of the tube joint.

b. The minimum acceptable increase in diameter as a percentage of initial tube wall thickness or interference fit for production.

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c. The maximum rate of increase in tube diameter after being expanded into the tubesheet to prevent discontinuity in the tube surface.

d. Recommendation for whether a limit should be established for a maximum inside diameter after a maintenance rerolling.

e. The maximum acceptable shop increase in diameter as a percentage of the initial tube wall thickness or interference fit for production control; this value is set to ensure the feasibility of rerolling the tubes several times (in the shop and in service for corrective maintenance).

f. The recommended increment of tube inside diameter change for maintenance rerolling.

g. The expander control settings for the production rolls, along with the instructions for accomplishing correct expansion.

A.3.1.2 Tensile test for rolled joints in type I, class 1 coolers. The tensile test for rolled joints in type I, class 1 coolers should be as follows:

a. A seven-tube mockup should be prepared to test the tensile strength of the thinner tubesheet joint on double tubesheet units. The mockup should consist of two tubesheet sections identical in fabrication and thickness to the thinner of the tubesheets and separated by a distance equal to the tubesheet separation on the production units. The pattern of the seven tube holes should be centered on the mockup tubesheets and should be otherwise identical to the production pattern. The total area of the tubesheet should be that which will produce the tensile load on the tubes when the hydrostatic pressure in the void space is at the pressure specified in 4.3.3.1. At least one of the tubesheets should be unrestrained, except for the tubes, so that the load imposed by the hydrostatic test will be borne by the tubes. The mockup should be prepared using the exact production tube rolling procedure and should be pressurized to the hydrostatic test pressure specified in 4.3.3.1.

b. Failure in the tube should occur before failure in the tube-to-tubesheet joint. Tube failure is considered to have occurred when the tube between the tubesheets either breaks or undergoes a tensile stress (due to pressure alone) equal to the minimum yield strength of the tube. Failure of the tube-to-tubesheet joint is considered to have occurred when the tube slips in the tubesheet or when the joint leaks.

A.3.1.3 Design adequacy of type I, class 1 coolers. For applications for which cyclic life requirements are established, an analysis demonstrating the adequacy of the design should be prepared. The contractor should prepare a design report for class 1 coolers. The report should be self-contained and in a format to facilitate independent review of its content; it is imperative, therefore, that it be simple to follow and free from ambiguity. The report should contain the following, at a minimum:

a. Description of design requirements, such as steady-state and transient pressure, temperature, and external loading.

b. The regions of the components that were analyzed.

c. The materials that were used and their mechanical properties.

d. A general description of the methods of analysis and assumptions.

e. List of reference sources.

f. Computer programs properly identified and described.

g. Copies of computer printouts (input and output).

h. Stresses tabulated for each area of investigation and compared to allowable stresses for all stress categories.

i. Satisfaction of cyclic requirements, including stress concentration factors used in the cyclic analysis.

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A.3.1.4 Tube bending procedure. The tube bending procedure details the tube bending method and demonstrates the effectiveness of the method through the production and subsequent destructive examination of test bends. The tube bending procedure should be prepared using MIL-STD-1627. When specified (see 6.2), the procedure should include details and approval requirements for each combination of tube size, gauge, fin geometry, material, tube manufacturer, and tube bending vendor proposed for use. Demonstration for approval should be by destructive examinations of minimum bend radius samples for wall thinning and roundness that should satisfy the requirements of ASTM B395/B395M. Bending radii for U-bend tubes should be not less than the dimensions specified in table XIII. The approved procedure should be referenced on the appropriate tube drawings.

A.3.1.5 Tube-to-tubesheet joint quality assurance plan. When specified (see 6.2), a quality assurance plan for the tube-to-tubesheet joint should be developed and submitted to the command or agency concerned for approval. The plan should cover the details of the manufacture and inspection process associated with completing the tube-to-tubesheet joint to provide high confidence that this joint will meet the cyclic life requirements specified (see 6.2) with no leakage. This plan should include the following, at a minimum:

a. Tooling and machining centers to be used.

b. Setup process, tolerance stack-up, and control.

c. Inspection measurement techniques and equipment and associated calibration requirements.

d. Methods used to preclude machining grooves out of position.

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GUIDANCE FOR ENGINEERING DRAWINGS TECHNICAL CONTENT

B.1 SCOPE

B.1.1 Scope. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only. This appendix provides procuring activities guidance on information that should be specified by the acquisition documents for inclusion on the drawings, when specified (see 6.2).

B.2 APPLICABLE DOCUMENTS

B.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

B.2.2 Government documents.

B.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-19521 - Anode Retaining Support Plugs and Anode Selection and Installation Design Criteria for Shipboard Condensers and Heat Exchangers

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-31000 - Technical Data Packages

(Copies of these documents are available online at https://quicksearch.dla.mil.)

B.2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

B.3 SUGGESTED ENGINEERING DRAWINGS CONTENTS

B.3.1 Drawing contents. Drawings should be in accordance with MIL-STD-31000. The cooler drawings should be those of the cooler contractor. Cooler drawings that are not prepared by the cooler contractor should be considered for preliminary review only. Assembly and detail drawings of the cooler, subassemblies, and parts should be level III drawings. Any proposed areas of departure from the specifications should be explicitly requested, with supporting basis, by the initial letter of submittal and should be called out as exceptions in the certification data sheet, “Statement of Conformance.”

B.3.1.1 Joints and threaded assemblies. Drawings should include recommended assembly bolting torques, including minimum and maximum values, and sequences or equivalent procedures for making up all joints and threaded assemblies. The type of thread lubricant to be used should be specified on the drawings (see 3.2.8.2).

B.3.1.2 Assembly drawings. Assembly drawings should show details of equipment furnished. Information usually furnished on outline or installation drawings may be shown on the assembly drawing. Sectional views and enlarged views should be adequate in number and quality employed to indicate details of parts, subassemblies, and assemblies. Where multiple detail views are shown on more than one sheet, the drawing titles should be distinctive. The title “Details” should not be sufficient.

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B.3.1.3 Certification data drawing. A certification data drawing should be furnished. It should be titled "Certification Data for …" followed by the kind of cooler, such as lubricating oil cooler, jacket water cooler, or battery cooler, and its designation by model number or by size and type, or the equivalent. The certification data drawing should be included in the first submission of ship equipment drawings.

B.3.1.3.1 Contents of certification data drawing. The certification data drawing should include the following information:

a. Identification by hull (side) number of the applicable ships; the hull numbers should be located above the title block.

b. Notes attesting to conformance to the specification, indicating the contract or order, the application (service) of the unit, and the number of coolers per ship and per unit of parent equipment (such as the engine). The following examples provide guidance for notes:

(1) NAVSHIPS contract N00024-7X-C1234 (this number is for example only) with ABC Engine Company; ABC Engine Company order 5678 on EFG Cooler Manufacturing Company.

(2) Service: Lubricating oil cooling for emergency diesel generator set or submarine main storage battery cooling.

(3) Eight coolers per ship, one cooler per blower; four coolers per ship, one cooler per engine; or 12 coolers furnished as stock material (use only when ship application is not known).

c. A drawing list tabulation. This list should include all equipment drawings that make up the given design. The revision symbols should be kept up-to-date to the time of manufacture so that it will finally indicate the latest revision of each drawing applicable to the equipment as built. The list should include the following three columns:

(1) Drawing title.

(2) Manufacturer's drawing number.

(3) Revision symbol.

d. A performance data tabulation to include the following:

(1) Identification of cooled and cooling mediums.

(2) Flow rate of cooled and cooling mediums in gallons per minute.

(3) Inlet temperature of cooled and cooling mediums in °F.

(4) Outlet temperature of cooled and cooling mediums in °F.

(5) Pressure drops of cooled and cooling mediums through cooler in lb/in^2.

(6) Velocities of cooled and cooling mediums at inlet connection in feet per second.

(7) Velocities of cooled and cooling mediums through tubes and shell in feet per second.

(8) Number of passes of cooled and cooling mediums through cooler.

(9) Design pressures of cooled and cooling mediums in lb/in^2 gauge. (For coolers for seawater service on submarines, no design pressure entry should be made for the seawater side.)

(10) Factory test pressure, shell-side in lb/in^2 gauge.

(11) Factory test pressure, tube-side in lb/in^2 gauge. (For coolers for seawater service on submarines, in lieu of an entry in the tabulation opposite this heading, the space should be left blank; the following procedure should be used instead. Beside the tabulation, use an arrow leading in to indicate this blank space, and at the tail of the arrow, add the following note: "For tube-side test pressure, see note … on … dwg … NAVSEA Nr … Test Data …". This note is to be completed when the information becomes available from the building yard.)

(12) Factory test pressure, space between double tubesheets in lb/in^2, where applicable. (For coolers for seawater service on submarines, in lieu of an entry in the tabulation opposite this heading, the space should be left blank; the following procedure should be used instead. Beside the tabulation, use an arrow leading in to indicate this blank space, and at the tail of the arrow, add the following note: "For test pressure for space between double tubesheets, see note … on … dwg … NAVSEA Nr … Test Data …".)

(13) Logarithmic mean temperature difference in °F.

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(14) Heat transfer rates for both service and clean tube conditions in Btu per hour per square foot per °F logarithmic mean temperature difference.

(15) Cooling surface in square feet. (For coolers using extended surface tubes, both the plain tube surface and the total surface [including fins] should be shown.)

(16) Heat transfer capacity at design point in Btu per hour.

(17) Dry weight and operating weight of cooler in pounds.

(18) Number of tubes, average effective tube length (inches), and tube size (inches). Specify if U-tubes are used.

e. A view of the identification plate with all data entered, except date of manufacture, serial number, and tube-side maximum test pressure for submarine seawater applications. For these applications, use an arrow leading in to indicate the blank spaces where tube-side maximum test pressure would have been entered. Place the notation, "Tube-side maximum test pressure not to be stamped on identification plate" at the tail of the arrow. This view should be enlarged as necessary to ensure that data entries meet lettering height requirements (⅛ inch).

f. The technical equipment manual number, provided that such manuals are required by the contract or purchase order and that the manufacturer has been apprised of the assigned number.

g. The component identification number allowance parts list (APL) number, if available.

h. Certifying signature of manufacturer's responsible engineering representative, whose professional qualification should be not less than that of a professional engineer.

i. An approval notation of the type required for type II drawings in accordance with MIL-STD-31000.

j. Compliance with MIL-A-19521 (see 3.3.14.3).

k. The title and number of the tube bending procedure used in the manufacture of the tube bundle, if applicable.

B.3.1.4 Contents of outline drawing. The outline drawing should include pull space for bundle removal, waterbox removal, waterbox bolting removal, and mounting bolt removal. It should also include the dry and wet weight of the cooler completely assembled, the location of the dry and wet centers of gravity, and the weight of each piece/part included in the assembly weighing more than 50 pounds.

B.3.1.5 Ultrasonic inspection. The requirement for ultrasonic inspection of forged or wrought material and studs, bolts, and nuts used in type I, class 1 coolers should be specified on the applicable cooler drawing. The requirement for ultrasonic inspection of tubesheets used in type I, class 4 coolers should be specified on the applicable cooler drawing.

B.3.1.6 U-bend finned tubes. Dimensional and ordering data for such tubes should be shown on applicable drawings.

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GUIDANCE FOR TECHNICAL MANUAL CONTENT

C.1 SCOPE

C.1.1 Scope. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only. This appendix provides procuring activities guidance on information that should be included in the technical manuals when specified (see 6.2).

C.2 SUGGESTED TECHNICAL MANUAL CONTENTS

C.2.1 Technical manuals. Technical manuals should be in accordance with C.2.1.1 through C.2.1.5.

C.2.1.1 Coolers with double tubesheets. Technical manuals should be furnished for coolers that have double tubesheets.

C.2.1.2 Submarine coolers subject to submergence pressure. Technical manuals should be furnished for submarine coolers subject to seawater submergence pressure. These manuals should include instructions as to the pre-stress values and corresponding torques to be applied to bolts in making up joints subject to submergence pressure and should also include the pattern of tightening the bolting. Technical manuals should give special attention to tube expanding procedures where single tube – double-tubesheet construction is involved.

C.2.1.3 Waterbox and tubesheet bolting. Technical manuals should contain a full description of the procedures for torqueing of bolts including minimum and maximum torque values.

C.2.1.4 Freshwater layup. Technical manuals should contain the process for accomplishing freshwater layup of the cooler.

C.2.1.5 Other coolers. For other coolers, technical manuals should be furnished only when specified.

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GUIDANCE FOR PRESERVATION AND PACKAGING

D.1 SCOPE

D.1.1 Scope. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only. This appendix provides procuring activities guidance on preservation and packaging of coolers.

D.2 APPLICABLE DOCUMENTS

D.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

D.2.2 Government documents.

D.2.2.1 Specifications and standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-P-645 - Paint, Alkyd Type, Formula Number 84

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2073-1 - Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil.)

D.2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D3951 - Standard Practice for Commercial Packaging

(Copies of this document are available online at www.astm.org.)

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 2 - Hand Tool Cleaning

(Copies of this document are available online at www.sspc.org.)

D.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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D.3 SUGGESTED PRESERVATION AND PACKAGING GUIDANCE

D.3.1 Preservation. Preservation should be level A or commercial, as specified (see 6.2).

D.3.1.1 Level A.

D.3.1.1.1 Cleaning and drying. Prior to preservation (unit protection), each cooler should be cleaned and dried by a process and procedure, respectively, in accordance with MIL-STD-2073-1, which will ensure removal, without damage to the cooler, of all corrosion, dirt, grease, rust, scale, and other foreign material.

D.3.1.1.2 Unit protection. Unit protection and methods of preservation of coolers should be in accordance with MIL-STD-2073-1 and D.3.1.1.2.1 through D.3.1.1.2.3.

D.3.1.1.2.1 Coolers constructed of corrosion-resistant materials. Method III should apply to coolers constructed of corrosion-resistant materials. Closure of openings should be specified in accordance with D.3.1.1.2.3.

D.3.1.1.2.2 Coolers constructed of ferrous or dissimilar materials. Method I should apply to coolers constructed of ferrous or dissimilar materials wherein corrosion or deterioration will occur. Unpainted exterior surfaces should be protected either by painting with a primer in accordance with TT-P-645 or coating with a non-tacky, hard-drying preservative in accordance with MIL-STD-2073-1. Use of a primer in accordance with TT-P-645 should include surface cleaning in accordance with SSPC-SP2 prior to application of a minimum of one coat of primer to a dry-film thickness of 2 to 4 mils per coat. The oil sides (lubricating or hydraulic) should be protected with a preservative in accordance with MIL-PRF-16173, grade 2 or with the normal operating oil, if specified (see 6.2). Unless otherwise specified (see 6.2), the seawater and freshwater sides of the coolers should be protected with a preservative in accordance with MIL-STD-2073-1. Closure openings should be in accordance with D.3.1.1.2.3.

D.3.1.1.2.3 Closure of openings. Openings such as inlets, outlets, and vents should be covered with a flexible, greaseproof, waterproof barrier material and a blank flange of mild steel, tempered hardboard, or fully waterproofed plywood. Mild steel flanges should be not less than 1/16 inch thick. For openings having a diameter up to and including 9 inches, 1/8-inch thick tempered hardboard should be used. For openings having a diameter over 9 inches, 1/4-inch thick tempered hardboard or fully waterproofed plywood should be used. Flanges should be secured with not less than four bolts, with nuts, and with lockwashers. On openings up to and including 2½ inches in diameter, plastic plugs or caps may be used in lieu of the barrier material and blank flanges.

D.3.1.2 Commercial. Commercial packaging (cleaning, preservation, cushioning, and unit pack) should be in accordance with ASTM D3951. Closure of openings should be in accordance with D.3.1.1.2.3.

D.3.1.2.1 Container modification. Shipping containers exceeding 200 pounds gross weight should be provided with a minimum of two 3-by-4-inch nominal wood skids laid flat or a skid- or sill-type base, which will support the material and facilitate handling by mechanical handling equipment during shipment, storage, and stowage.

D.3.2 Packing. Packing should be level A, B, C, or commercial as specified (see 6.2).

D.3.2.1 Navy fire-retardant treated lumber and plywood. When specified (see 6.2), all lumber and plywood (including laminated veneer material used in shipping containers and pallet construction) should be fire-retardant treated material in accordance with MIL-L-19140 as specified in table D-I.

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TABLE D-I. Fire-retardant treatment for levels A, B, and C.

Level	Fire-retardant treatment
A and B	Type II – weather resistant
	Category 1 – general use
C	Type I – non-weather resistant
	Category 1 – general use

D.3.2.2 Levels A, B, and C. Each cooler should be anchored, blocked, braced, and cushioned with its shipping container in accordance with the applicable container specification or appendix. Shipping containers should be of wood-cleated box construction or wood or plywood, open- or closed-type crate construction in accordance with MIL-STD-2073-1. Boxes exceeding a gross weight of 200 pounds should be modified with skids in accordance with the applicable box specification. Crates should be used for coolers exceeding the weight limitation of the box specification. When shipped in open-type crates, coolers should be provided with flexible, reinforced, waterproof barrier material shrouds. Plastic shrouds should be a minimum of 0.006 inch thick. Shrouds should be secured to prevent their loss or damage during handling, shipment, and storage. Shipping container closure and reinforcing should be in accordance with MIL-STD-2073-1.

D.3.2.3 Commercial. Commercial packing should be in accordance with ASTM D3951 and herein. Shipping containers exceeding 200 pounds gross weight should be provided with a minimum of two 3-by-4-inch nominal wood skids laid flat, or a skid- or sill-type base, which will support the material and facilitate handling by mechanical handling equipment during shipment, storage, and stowage.

D.3.3 Levels A, B, C, and commercial marking. In addition to any special marking specified (see 6.2) and herein, shipping containers should be marked in accordance with MIL-STD-2073-1.

D.3.3.1 Special marking.

D.3.3.1.1 Equipment serial number. The equipment serial number should be included with the exterior identification and markings.

D.3.3.1.2 Open crates. Coolers packed in open crates exceeding 1000 pounds gross weight should be marked on one side and one end with the following:

DO NOT STACK MORE THAN THREE HIGH

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CONCLUDING MATERIAL

Preparing activity:
Navy − SH
(Project 4420-2023-001)

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